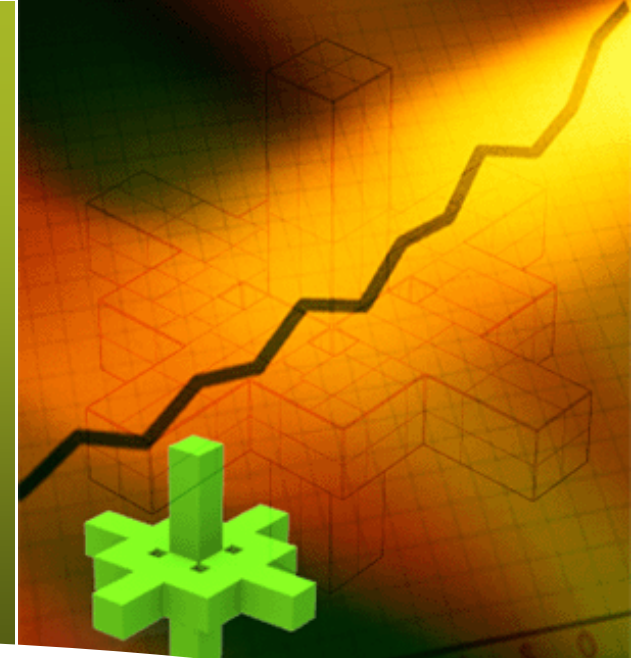
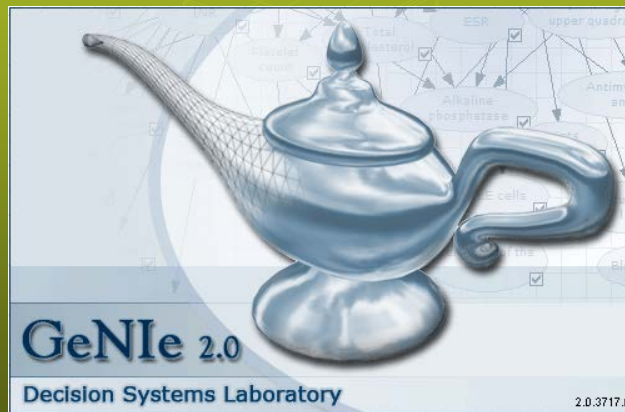


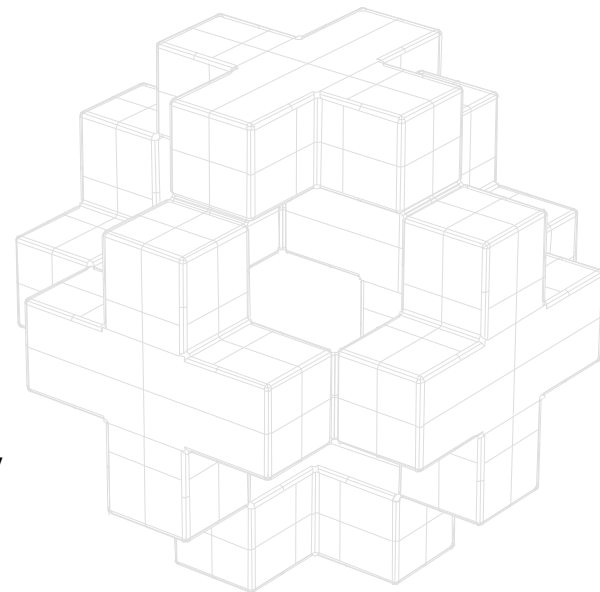


Bayesian Networks Practice

베이지안망 연습

2016-03-17

Byoung-Hee Kim
Biointelligence Lab, CSE,
Seoul National University





Agenda

- ❖ **Motivating Example** (15 min)
- ❖ **Bayesian Network & Probabilistic Graphical Model** (10 min)
- ❖ **Tool for Practice: GeNIe** (5 min)
- ❖ **Bayesian Network Practice with GeNIe & Weka**
 - Main Practices
 - Designing a knowledge system (model) (20 min)
 - Inference with the given model (30 min)
 - Learning Bayesian Network models from data (30 min)
 - Homework & Exercise (20 min)
- ❖ **Appendix**

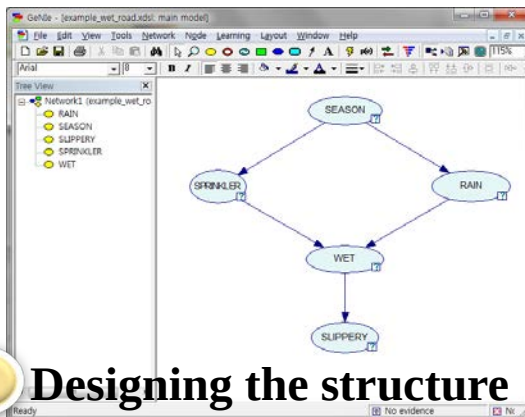
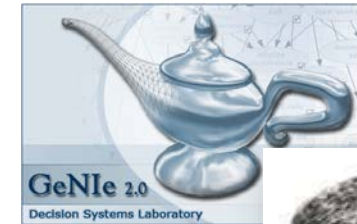
Practice Outline

1 Designing a Bayesian Network

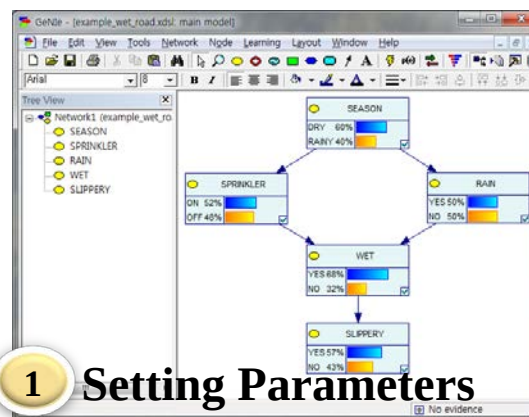
- Tool: GeNIe (<https://dslpitt.org/genie/>, <http://www.bayesfusion.com/>)

2 Inference based with a Bayesian Network model

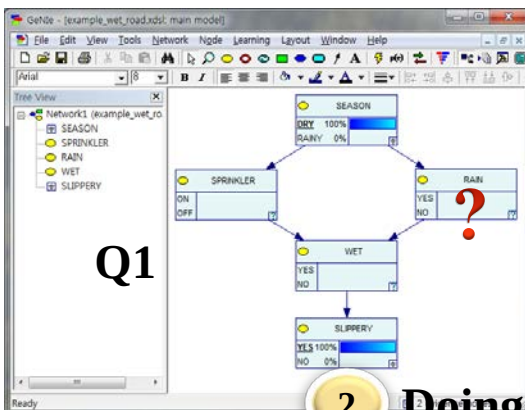
3 Learning Bayesian Network Models from data



1 Designing the structure

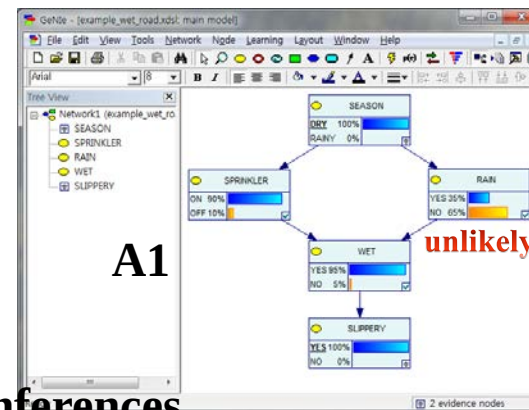


1 Setting Parameters



Q1

2 Doing inferences

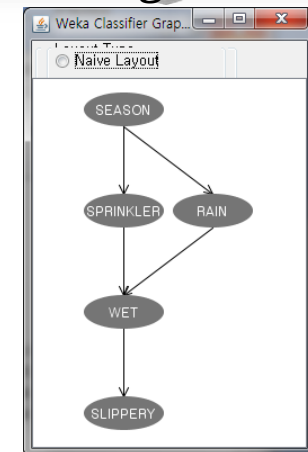


A1

unlikely

	A	B	C	D	E
1	SEASON	SPRINKLER	RAIN	WET	SLIPPERY
2	RAINY	OFF	NO	NO	NO
3	RAINY	OFF	YES	NO	NO
4	DRY	ON	NO	YES	NO
5	RAINY	OFF	YES	YES	YES
6	DRY	OFF	YES	NO	NO
7	RAINY	OFF	YES	NO	NO
8	DRY	ON	NO	YES	YES
9	DRY	ON	NO	YES	NO
10	DRY	OFF	YES	YES	YES

3 Learning from data





Motivating Example

Why do we need graphs?

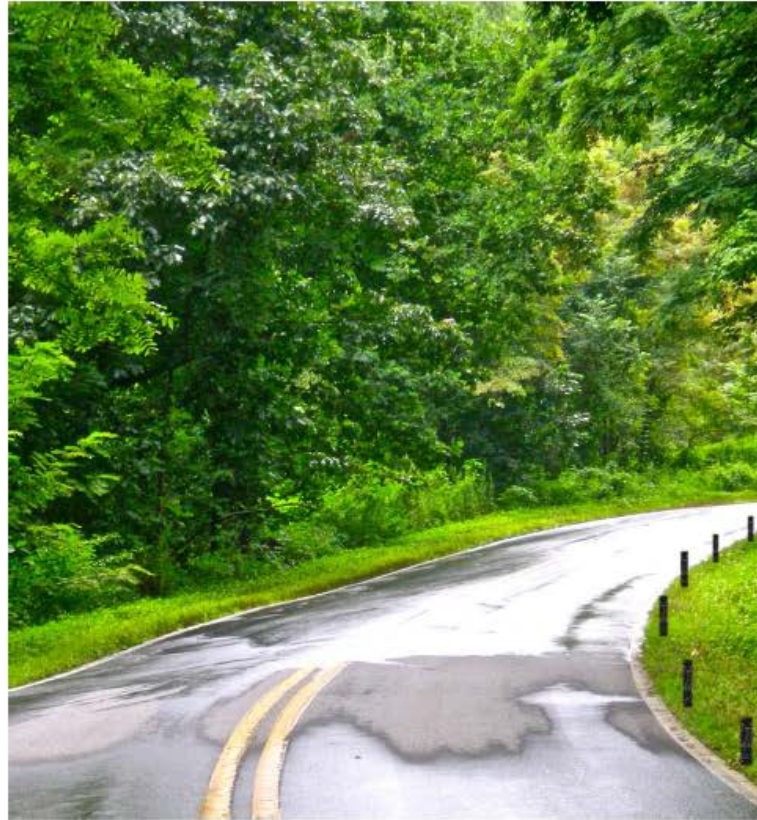


Figure: Motivating Example

Introduction

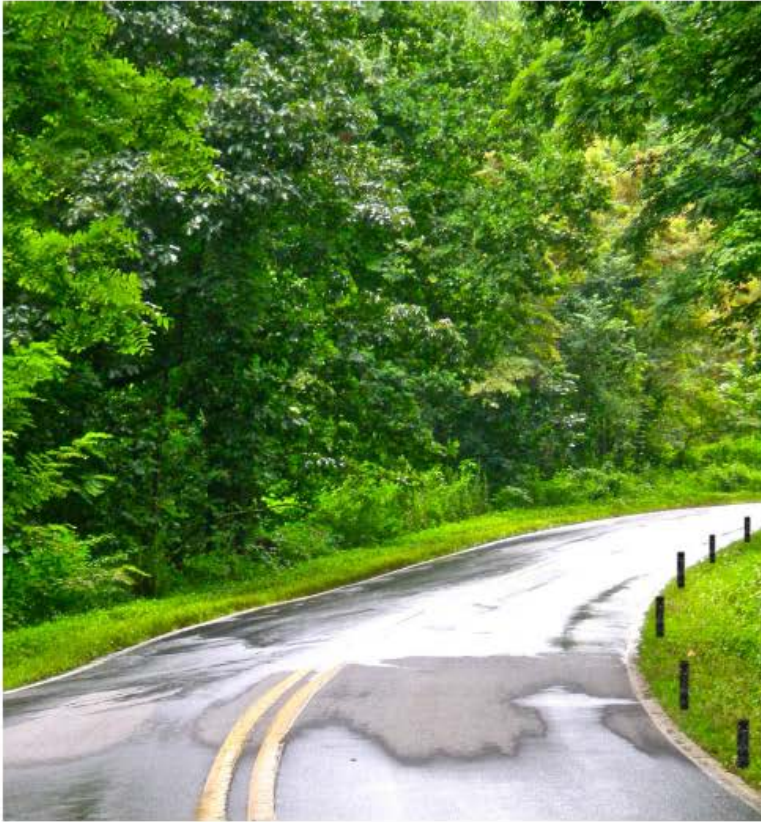


Figure: Motivating Example

Variables in the study:

- ▶ Season
- ▶ Sprinkler
- ▶ Rain
- ▶ Wetness of pavement(Wet)
- ▶ Slipperiness of pavement(Slippery)

Assuming binary states for all the variables

Ex) Season: dry or rainy

Ex) Sprinkler: ON or OFF

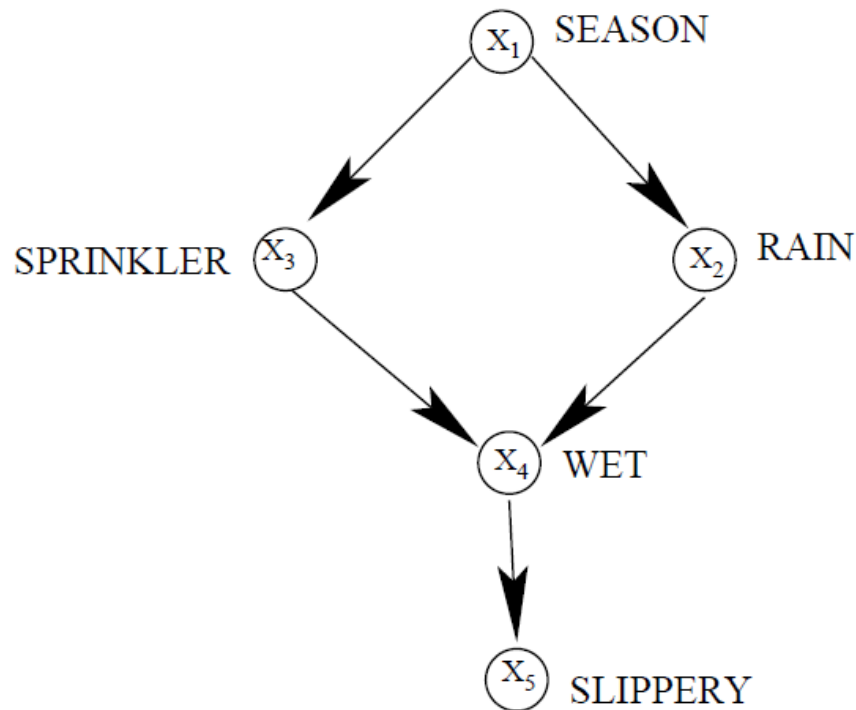
Introduction



# Variables	Table size
5	32
6	64
7	128
8	256
9	512
10	1,024
20	1,048,576
30	1,073,741,824

Figure: Motivating Example

Introduction



Conditional Probability Distributions

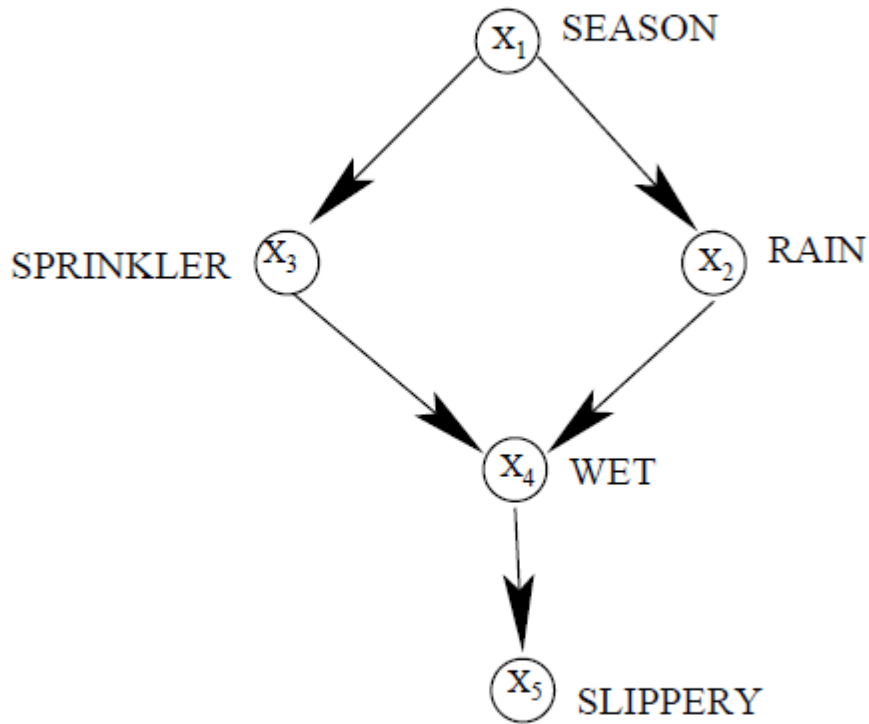
- ▶ $P(X_1) : 2$
- ▶ $P(X_3|X_1) : 4$
- ▶ $P(X_2|X_1) : 4$
- ▶ $P(X_4|X_2, X_3) : 8$
- ▶ $P(X_5|X_4) : 4$

Total # of Table Entries = 22

Figure: DAG Representation
(directed acyclic graph)



Markovian Parents



Markovian Parents

$X_1 : \phi$

$X_2 : \{X_1\}$

$X_3 : \{X_1\}$

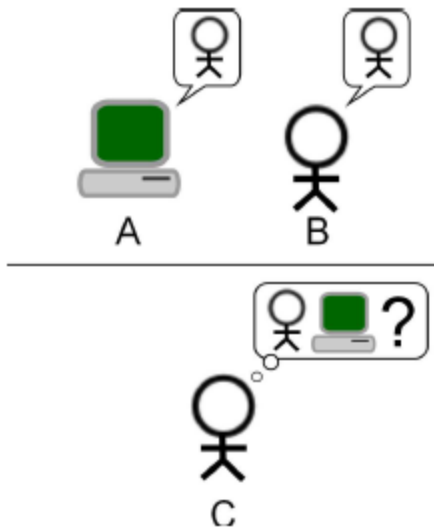
$X_4 : \{X_2, X_3\}$

$X_5 : \{X_4\}$

Figure: Bayesian Network
representing dependencies

$$P(x_1, x_2, x_3, x_4, x_5) = P(x_1)P(x_2|x_1)P(x_3|x_1)P(x_4|x_2, x_3)P(x_5|x_4)$$

A “Mini” Turing Test in Causal Conversation



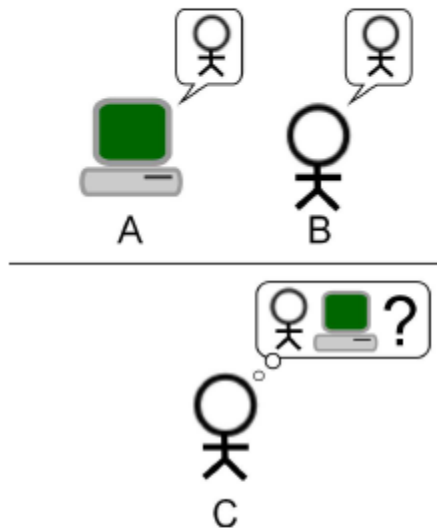
Input: Story

Question: What if? What is? Why?

Answers: I believe that...

Figure: Turing Test

A “Mini” Turing Test in Causal Conversation



The Story

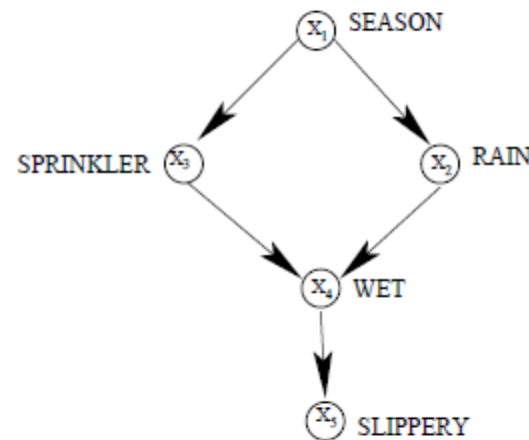
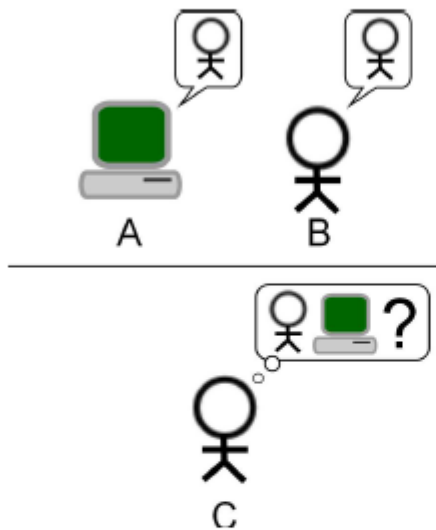


Figure: Turing Test

Q1: If the season is dry and the pavement is slippery, did it rain?

A1: Unlikely, it is more likely that the sprinkler was ON with a very slight possibility that it is not even wet.

A “Mini” Turing Test in Causal Conversation



The Story

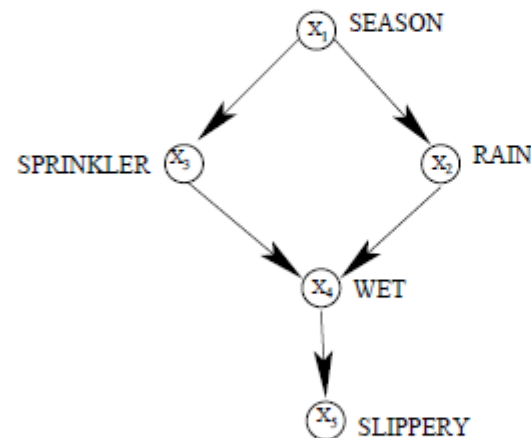


Figure: Turing Test

Q2: But what if we see that the sprinkler is OFF?

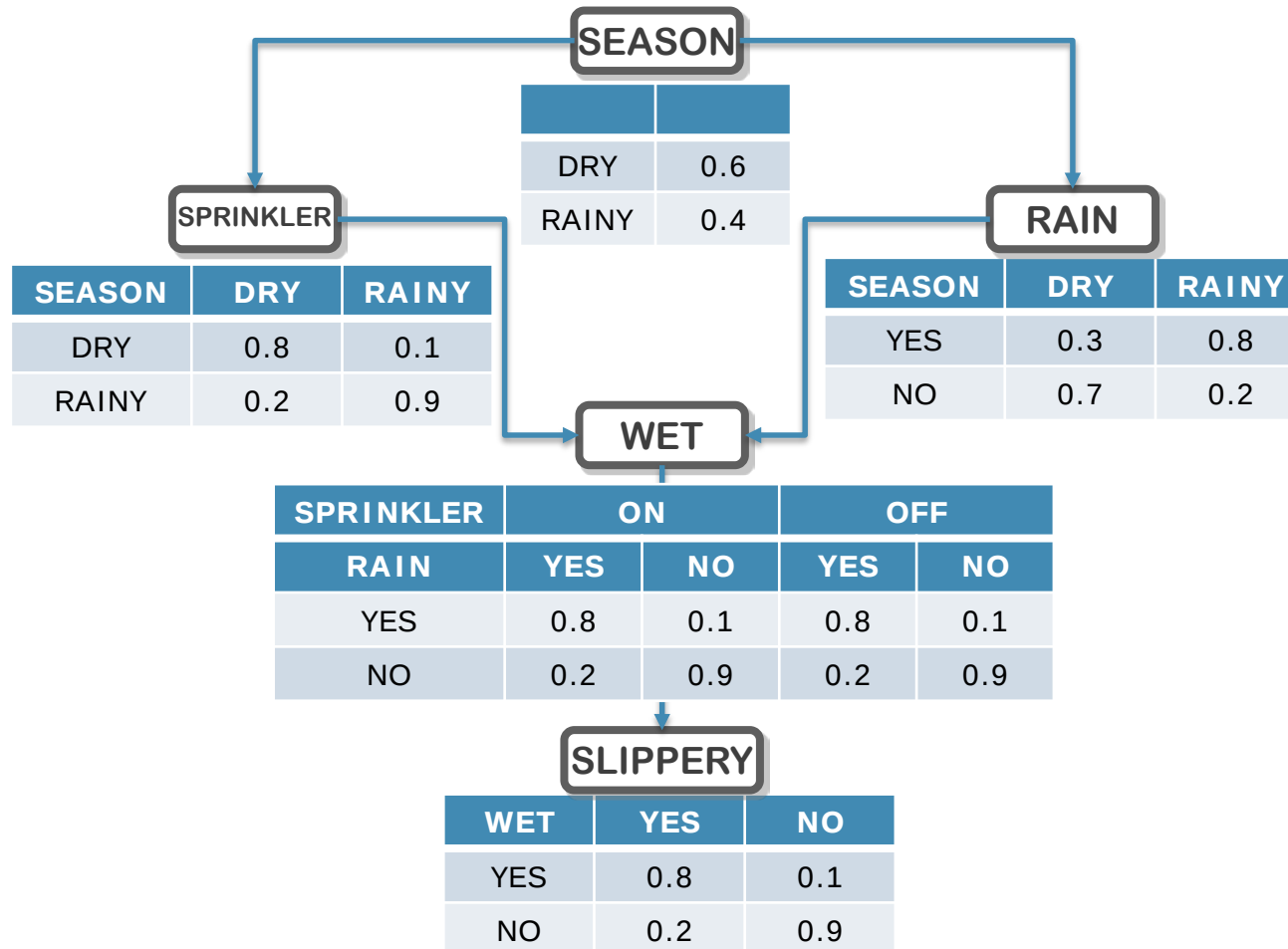
A2: Then it is more likely that it rained.

Without graphs, # of Table Entries = 32

Example: Knowledge Engineering

❖ Designing a Bayesian Network: Wet Road

- My own design of conditional probability tables

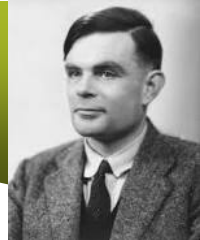




General Concept of Bayesian Networks & Probabilistic Graphical Models



ACM Turing Award



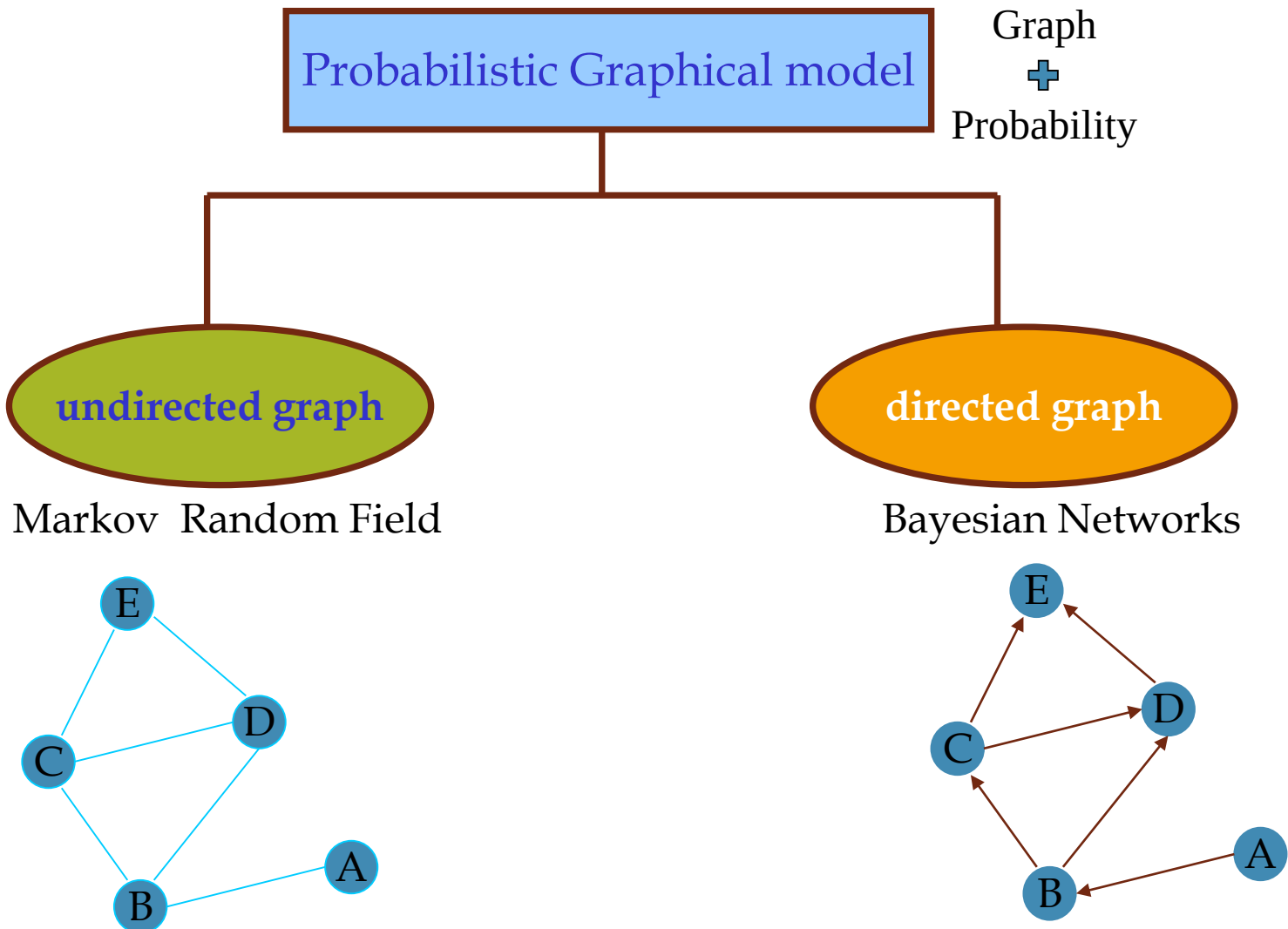
❖ Nobel Prize in Computing

❖ 2011 Winner: Judea Pearl (UCLA)



- For fundamental contributions to **artificial intelligence** through the development of a calculus for **probabilistic and causal reasoning**
- Invention of **Bayesian networks**
 - Pearl's accomplishments have “redefined the term ‘thinking machine’” over the past 30 years
 - BN mimics “the neural activities of the human brain, constantly exchanging messages without benefit of a supervisor”

Bayesian Network as a Probabilistic Graphical Model

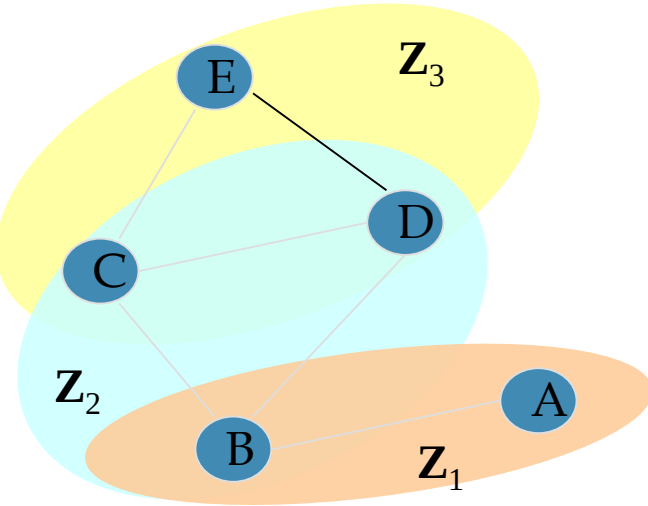


Representation of Joint Probability

The joint probability is effectively represented as the product of independent factors

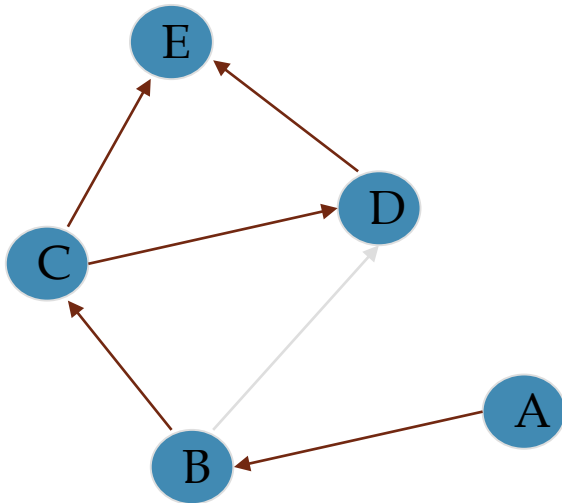
Markov Random Field

$$\begin{aligned} P(A, B, C, D, E) &= \frac{1}{\alpha} \prod_i \phi_i(Z_i) \\ &= \frac{1}{\alpha} \phi_1(A, B) \phi_2(B, C, D) \phi_3(C, D, E) \end{aligned}$$



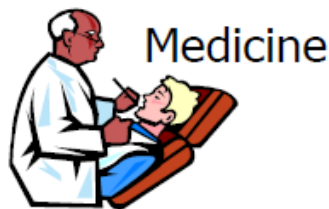
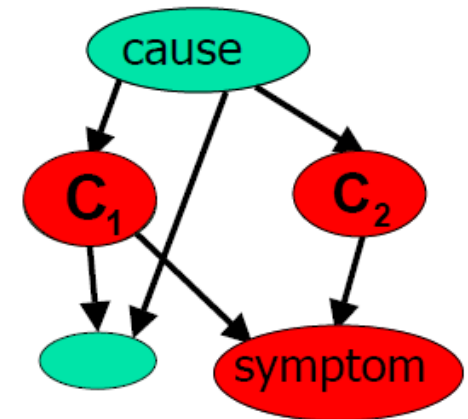
Bayesian Networks

$$\begin{aligned} P(A, B, C, D, E) &= P(A | \mathbf{Pa}(A)) P(B | \mathbf{Pa}(B)) P(C | \mathbf{Pa}(C)) \\ &\quad P(D | \mathbf{Pa}(D)) P(E | \mathbf{Pa}(E)) \end{aligned}$$

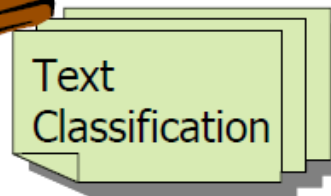


What are BNs useful for?

- Diagnosis: $P(\text{cause}|\text{symptom})=?$
- Prediction: $P(\text{symptom}|\text{cause})=?$
- Classification: $\max_{\text{class}} P(\text{class}|\text{data})$
- Decision-making (given a cost function)



Speech
recognition



Computer
troubleshooting



Microsoft Office Assistant

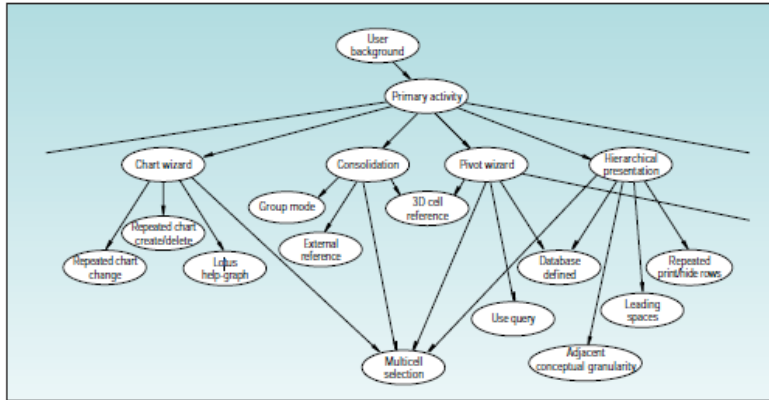


Figure 2. A portion of the Lumière Bayesian network user model.

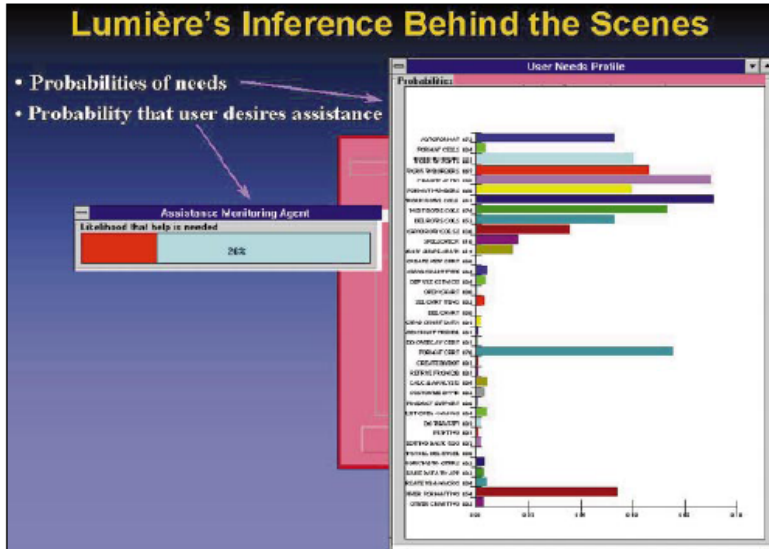
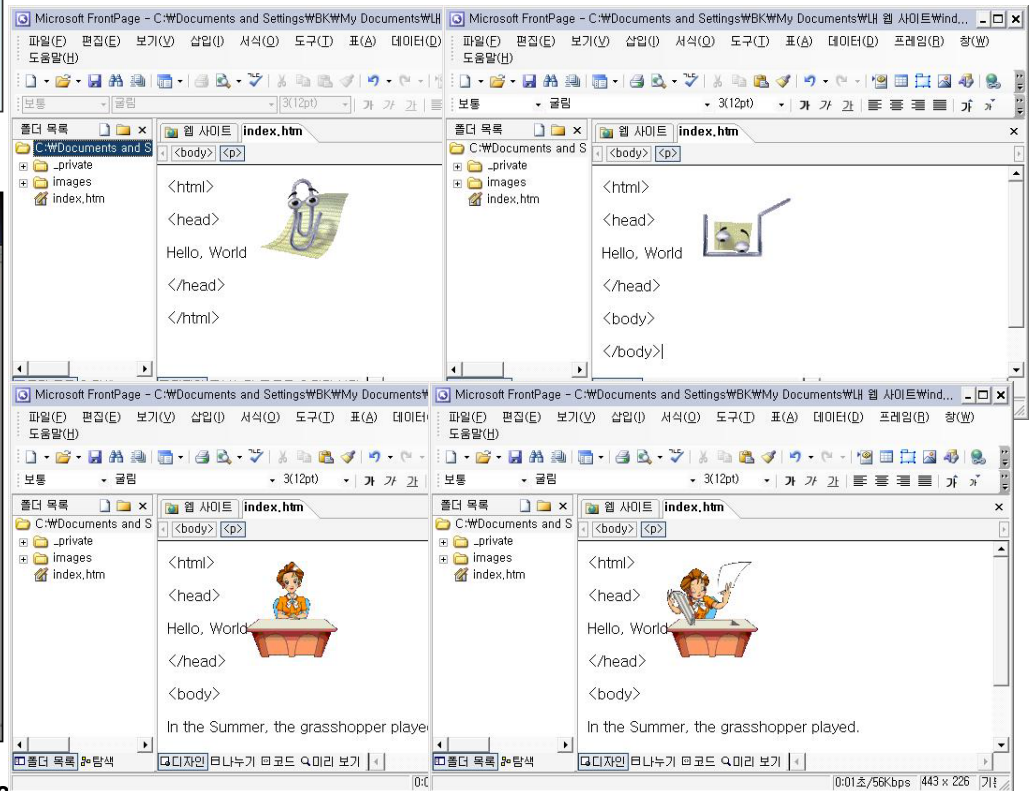
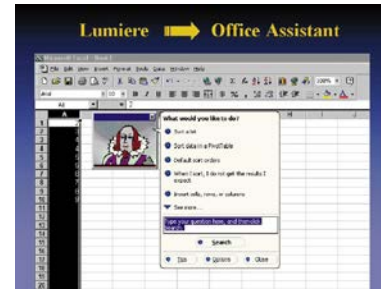
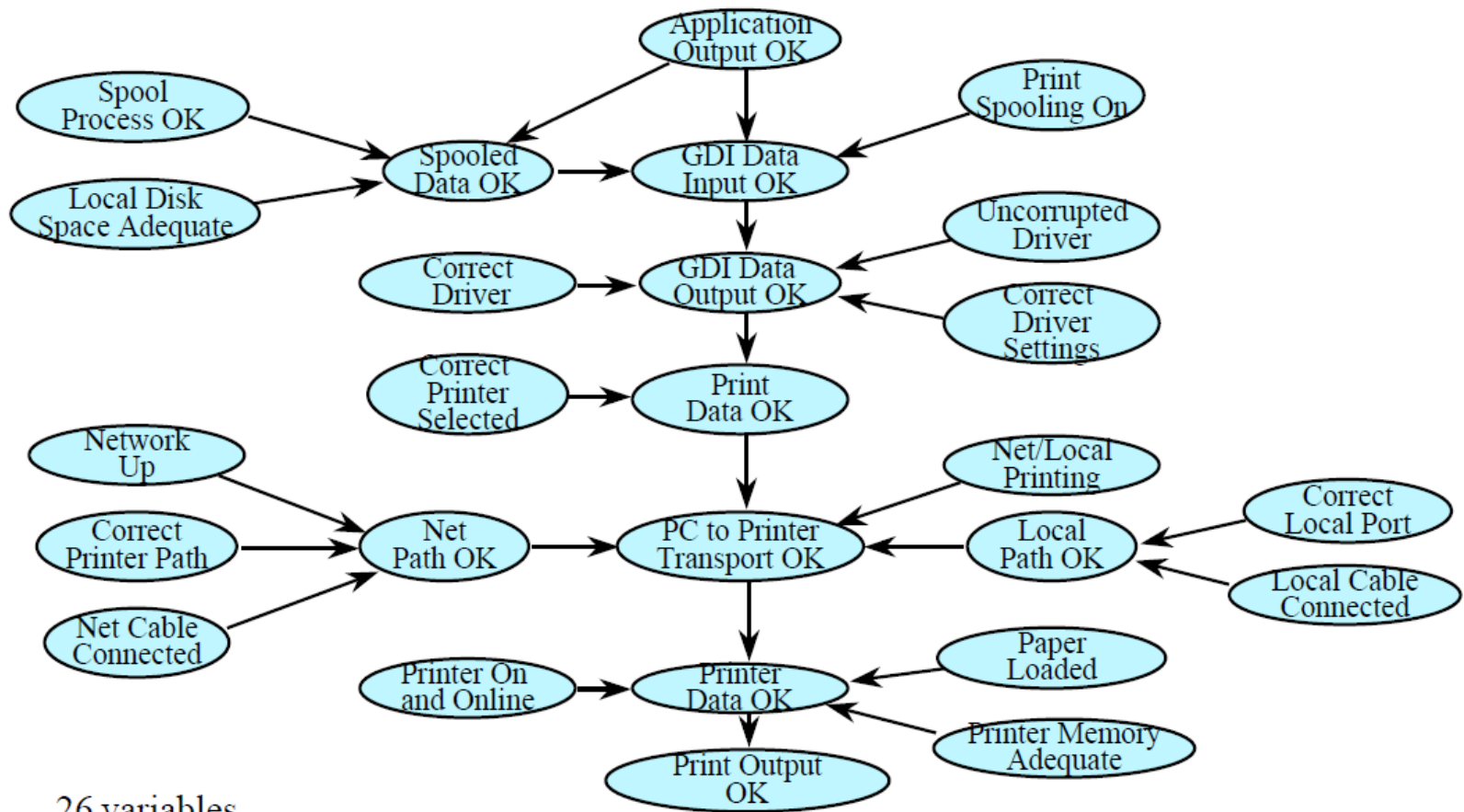


Figure 3. Lumière's Inference Behind the Scenes.



Example: Printer Troubleshooting



26 variables

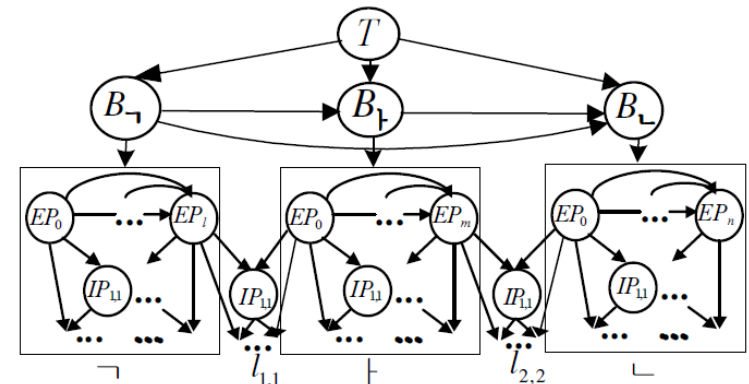
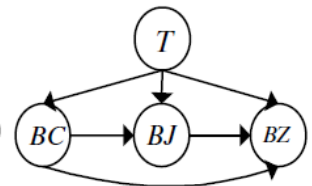
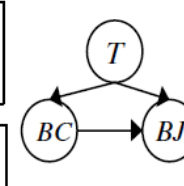
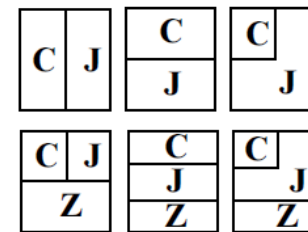
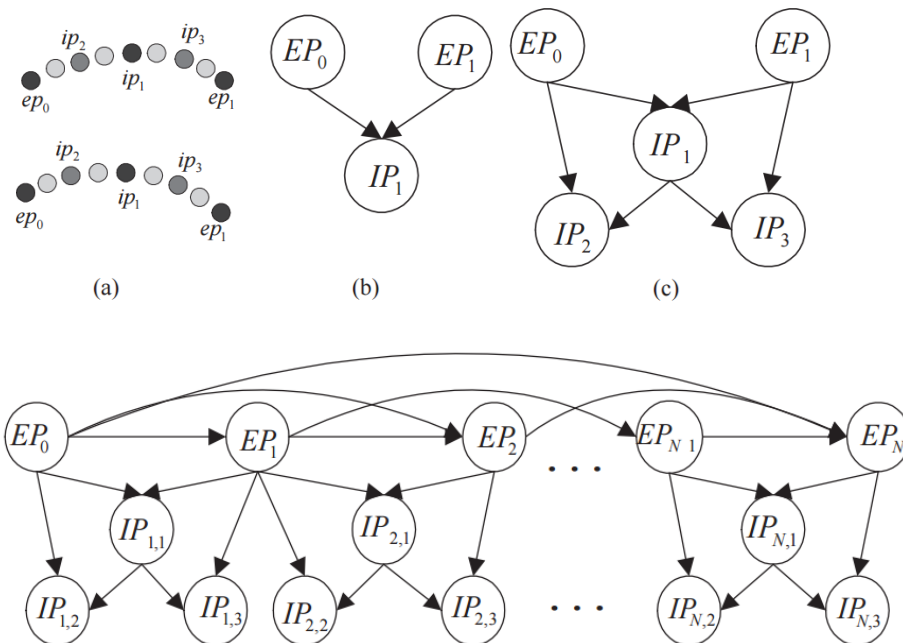
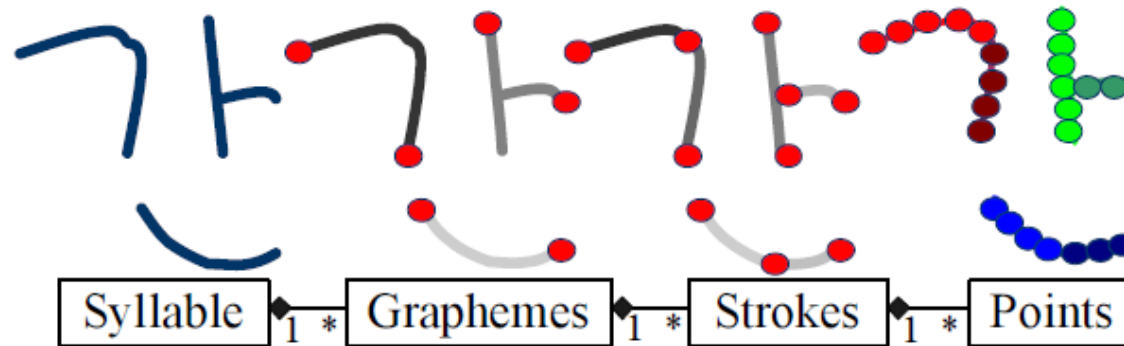
Instead of 2^{26} parameters we get

$$99 = 17 \times 1 + 1 \times 2^1 + 2 \times 2^2 + 3 \times 2^3 + 3 \times 2^4$$

[Heckerman, 95]

Example: Handwriting Recognition

[S.-J. Cho, 2003, 2004]





Applications Examples

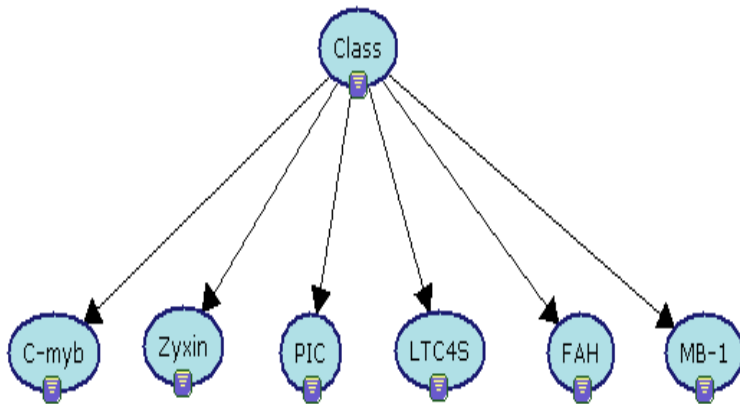
- ❖ APRI system developed at AT&T Bell Labs
 - Fraud/Uncollectible debt detection
 - learns & uses Bayesian networks from data to identify customers liable to default on bill payments

- ❖ NASA Vista system
 - predict failures in propulsion systems
 - considers time criticality & suggests highest utility action
 - dynamically decide what information to show

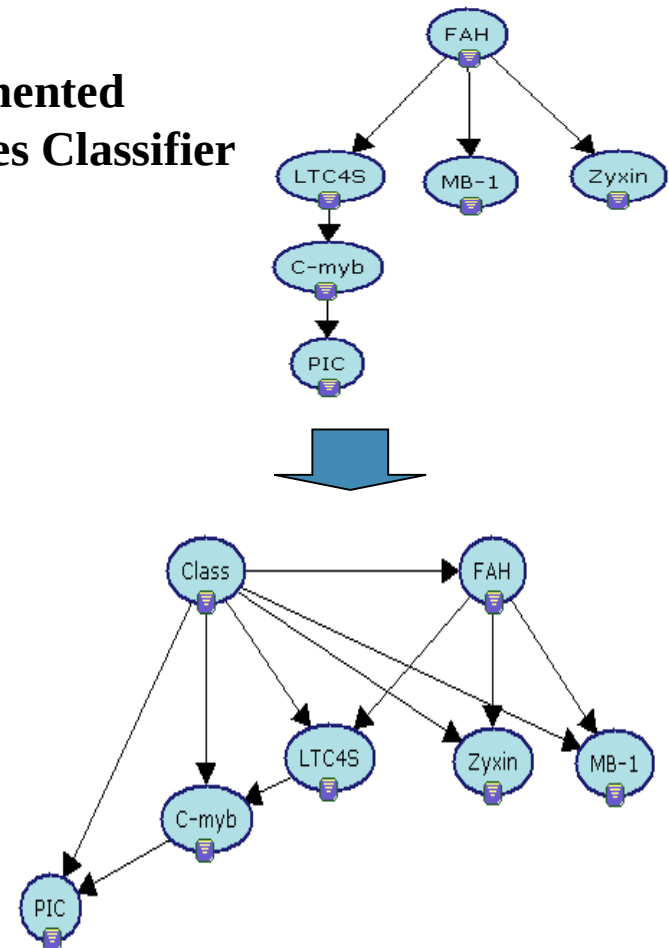
- ❖ Microsoft Pregnancy and Child-Care
 - Available on MSN in Health section
 - Frequently occurring children's symptoms are linked to expert modules that repeatedly ask parents relevant questions
 - Asks next best question based on provided information
 - Presents articles that are deemed relevant based on information provided

Bayesian Network Classifiers

Naïve Bayes Classifier



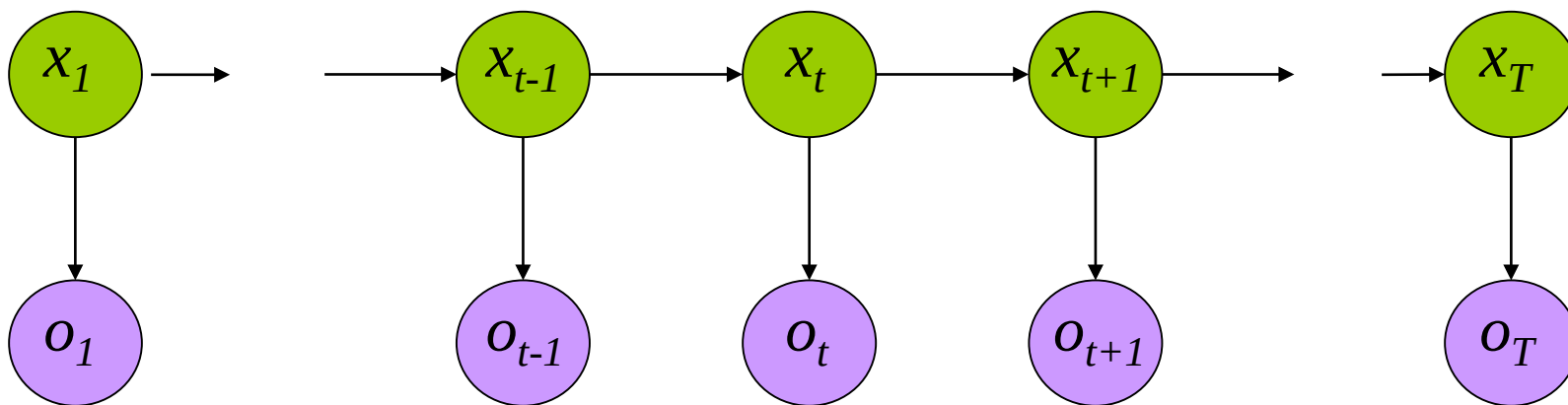
Tree-Augmented Naïve Bayes Classifier (TAN)



Bayesian Network as a Framework

❖ Many probabilistic models can be represented as Bayesian networks

- Hidden Markov Models
- Kalman Filters / Particle Filters
- PCA, ICA, ...
- Hierarchical Bayes models
- Etc.



❖ TakeHeart II: Decision support system for clinical cardiovascular risk assessment

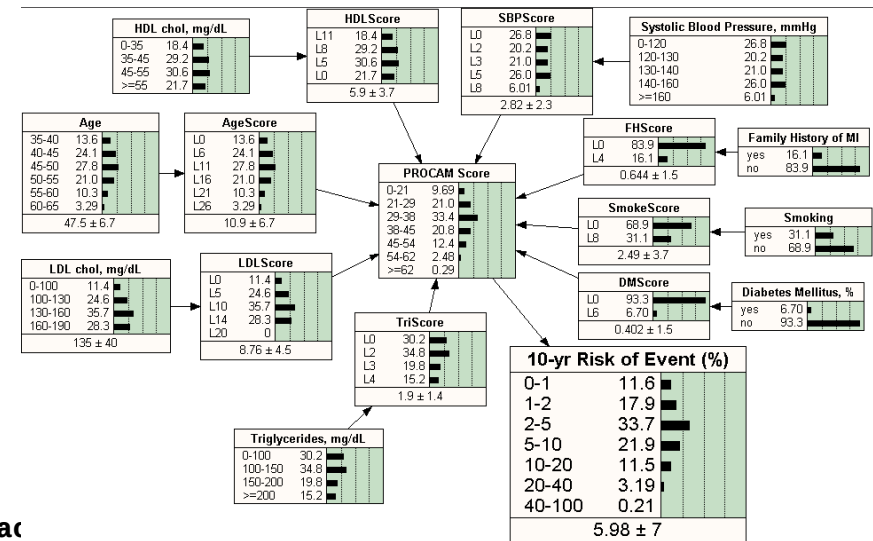
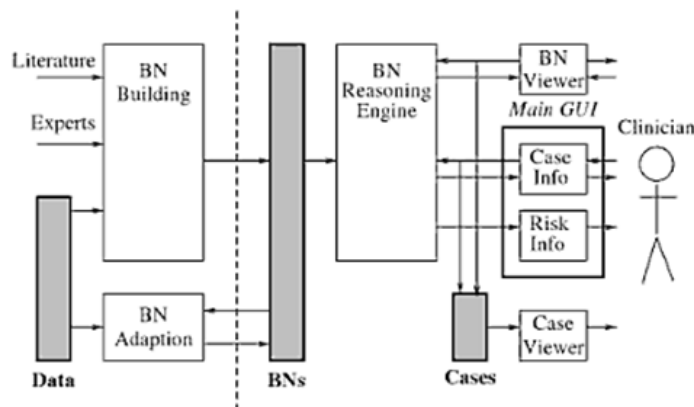
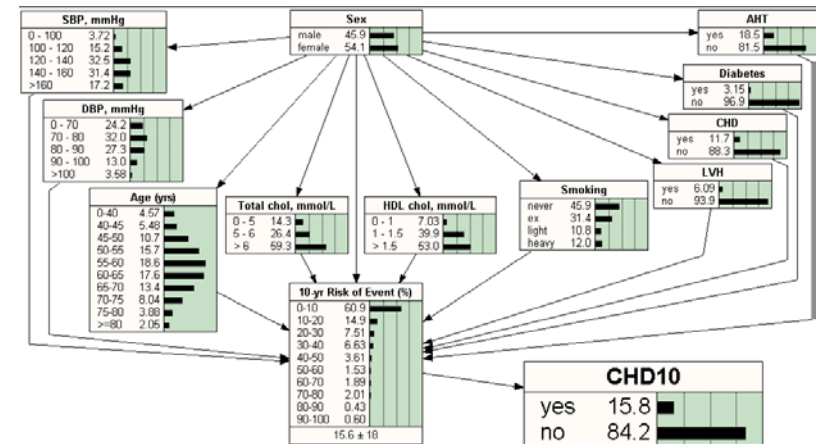
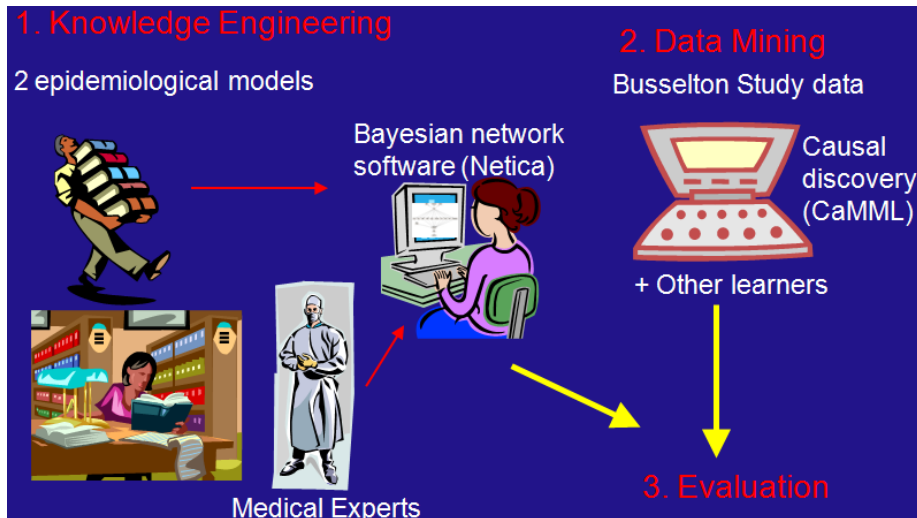
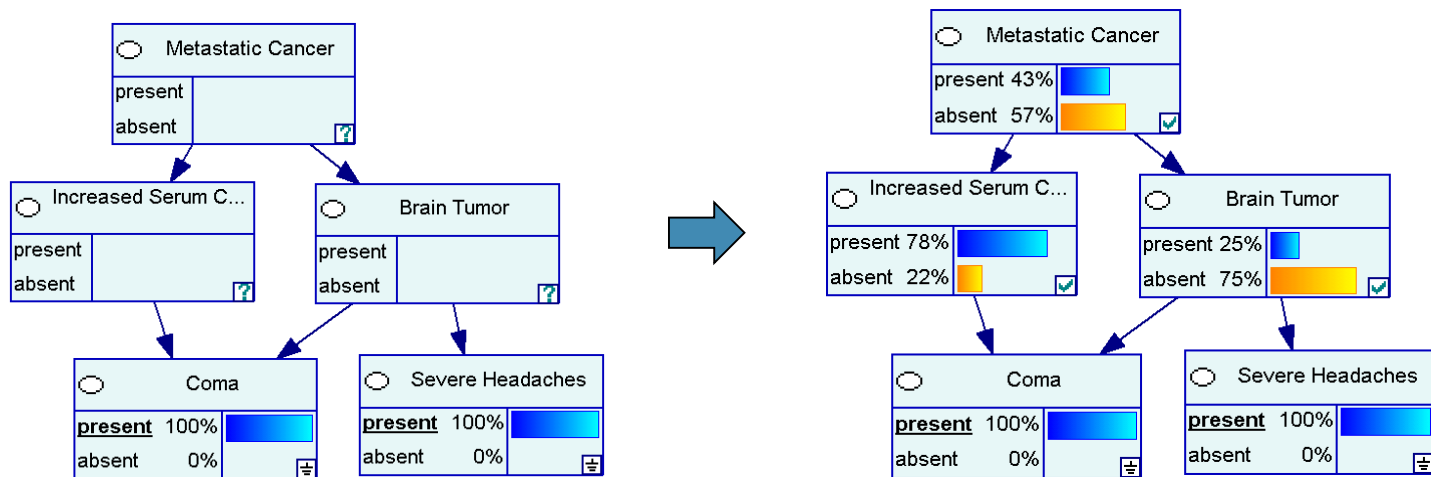
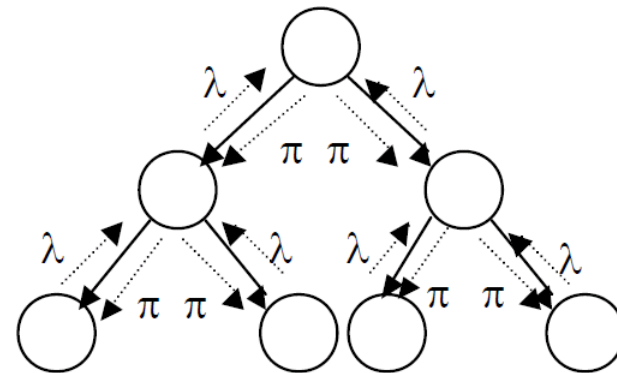


Figure 3.6 TakeHeartII Architecture: Construction and Adaptation (left) provide BNs used for risk assessment in a clinical setting (right).

- ❖ Given an assignment of a subset of variables (evidence) in a BN, estimate the posterior distribution over another subset of unobserved variables of interest.



Inferences viewed as
message passing along the network



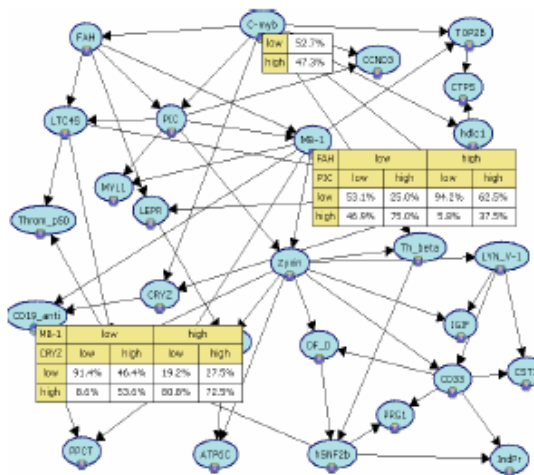
[illegible]

Type	AL	AL	AL	AL	AL	AL	AL	AL
Control	high	low	high	high	high	low	high	high
RAE-Rose	low	low	low	low	low	low	low	high
PROFESS	low	low	high	high	high	high	high	high
LeakDetector	low	low	low	low	low	low	low	high
MB-1 test	high	low	low	high	high	low	high	high
Qubit	low	low	low	low	low	low	low	high
SDMR-CL	high	low	high	high	high	high	high	high
NIR-Ves	low	low	low	low	low	low	low	low
BETMOSS	high	high	high	high	high	high	high	low
CO3D CLD	low	low	low	low	low	low	low	low
CR2/CO	low	low	high	low	high	high	high	high
DE-Ducan	low	low	low	low	low	low	low	low
MIL-1 test	high	low	high	high	high	low	low	low
APR-act	low	low	low	high	low	low	low	high
Thermostat	high	low	high	high	high	high	high	high
Gal 14F	low	low	low	low	low	low	high	low
Transcript	low	low	high	low	low	high	low	low
Use nPM	low	high	low	high	low	low	low	low



Prior knowledge

BN = Structure +
Local probability
distribution



Bayesian network learning

- * Structure Search
- * Score Metric
- * Parameter Learning



Practice

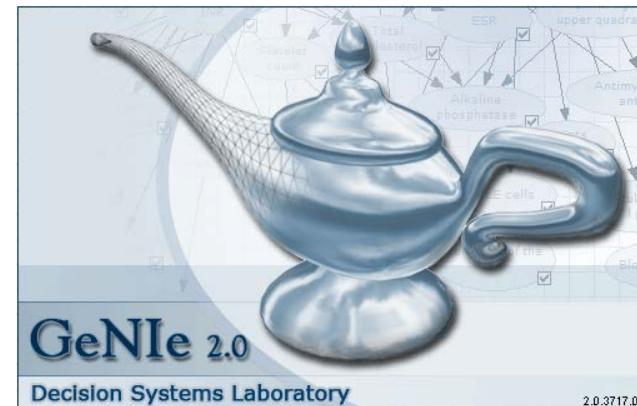
Bayesian Network – The Tool for Practice

❖ **GeNie:** <http://genie.sis.pitt.edu/>, <http://www.bayesfusion.com/>

- **GeNie** (Graphical Network Interface) is the graphical interface to SMILE, a fully portable Bayesian inference engine
- What you can do with **GeNie** are
 - Create and modify network models with a graphical editor
 - Building Bayesian networks, influence diagrams
 - Learning structures and parameters of Bayesian networks from data
 - Probabilistic inference with given Bayesian network

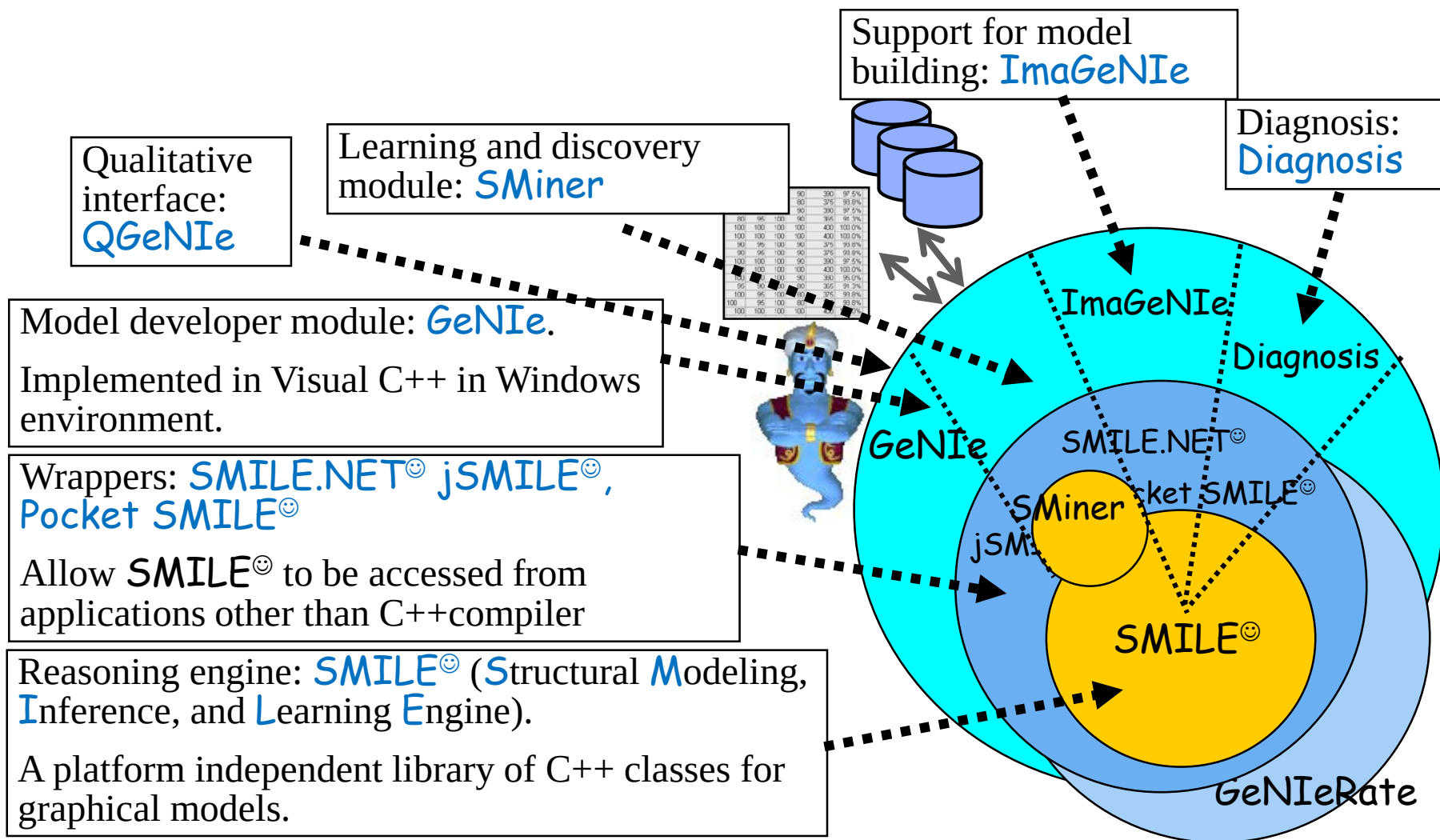
Note: **SMILE** (Structural Modeling, Inference, and Learning Engine) is a fully portable library of C++ classes implementing graphical decision-theoretic methods

(Look slide 42 for more information)



GeNIe and SMILE[☺]

A developer's environment for graphical decision models



❖ Designing a Disease-Test Model

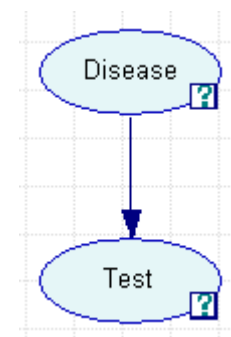
- Building a Bayesian network manually using GeNIe

❖ Inference

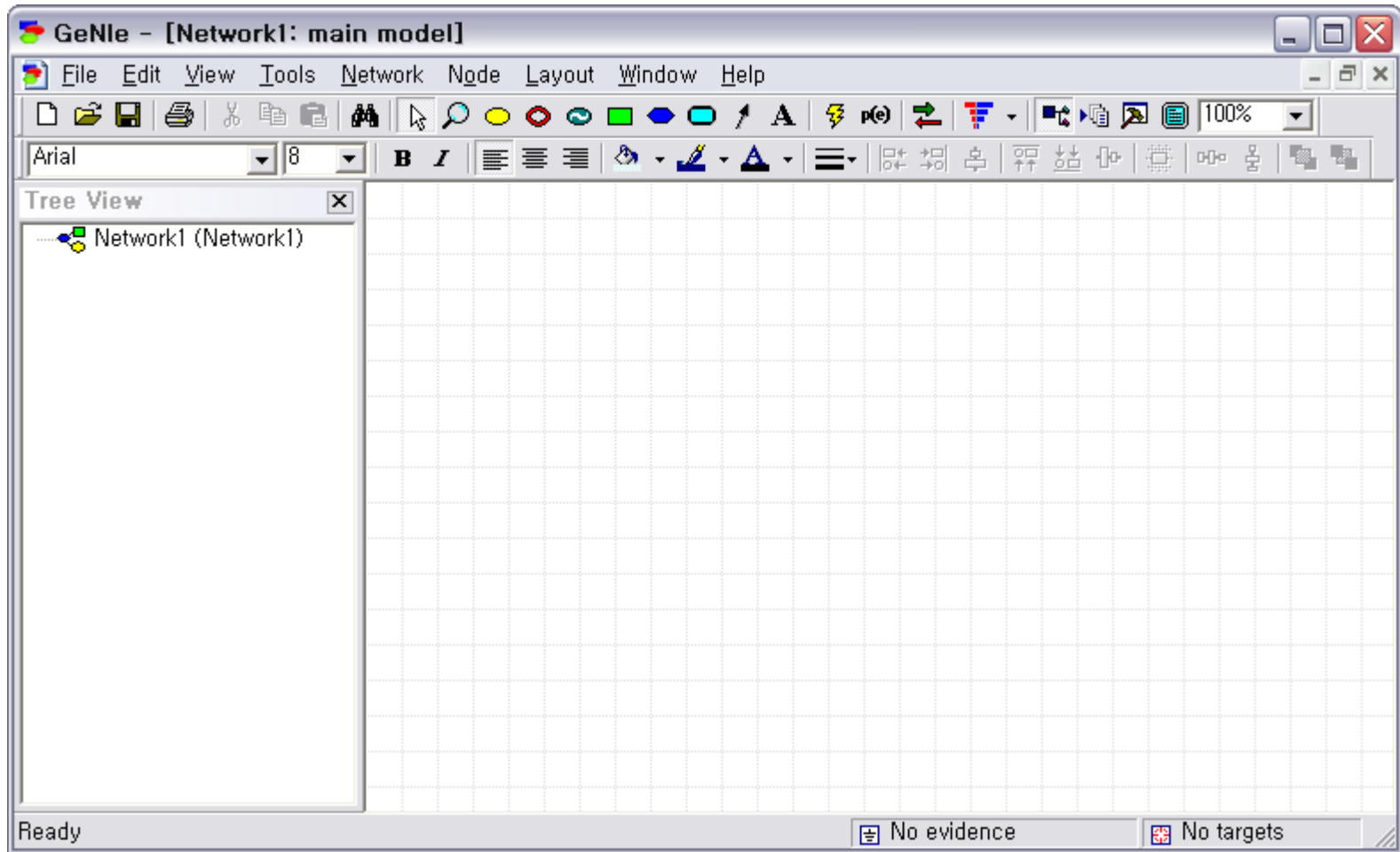
- In GeNIe, try following menus
 - 'Network – Update Beliefs', or 'F5'
 - Network - Probability of Evidence
 - Network - Annealed MAP

❖ Learning from Data

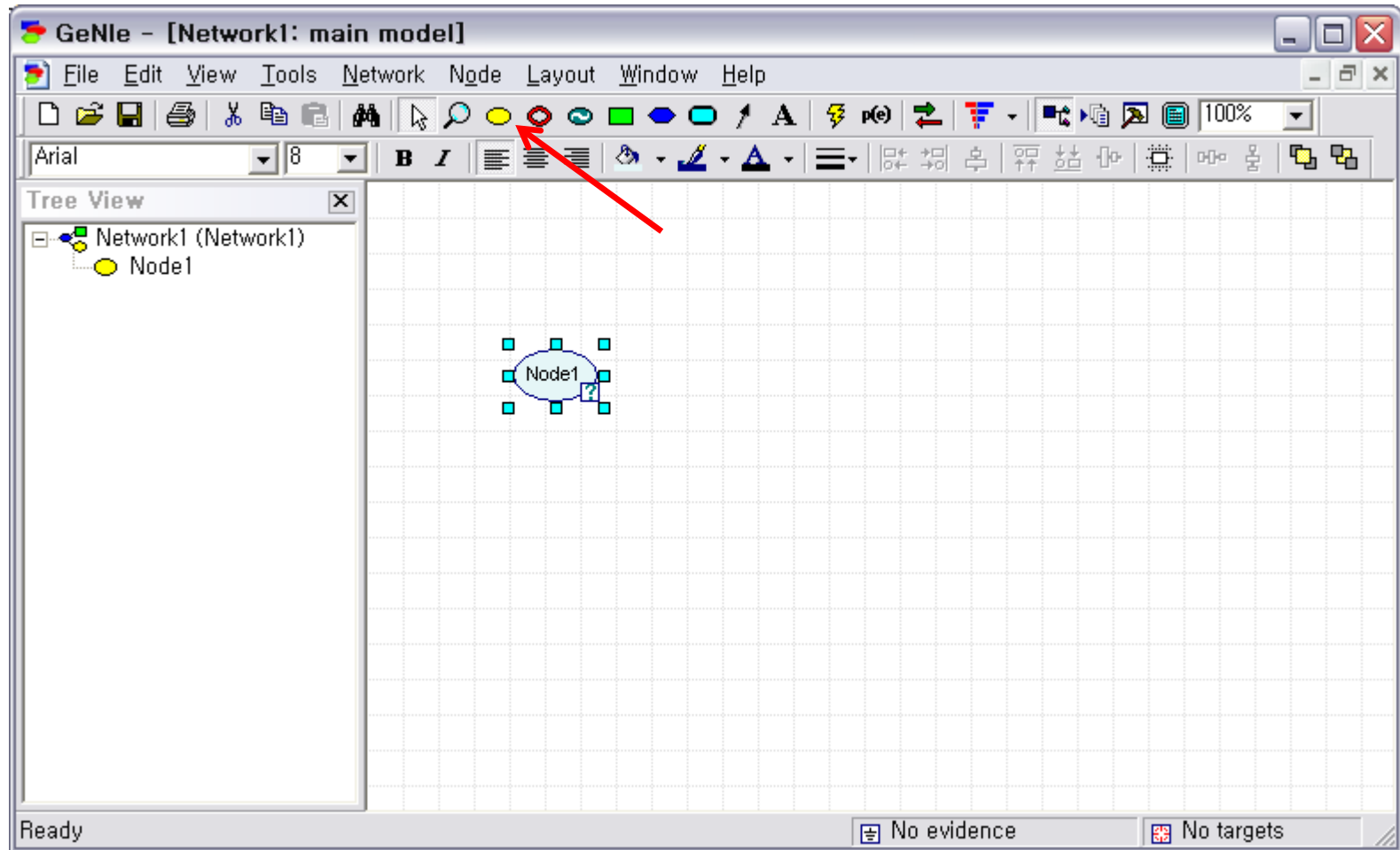
- (Data generation using an existing BN model)
- Learning Bayesian network from data using GeNIe
 - Structure learning
 - Parameter learning
- Analyzing the resulting model



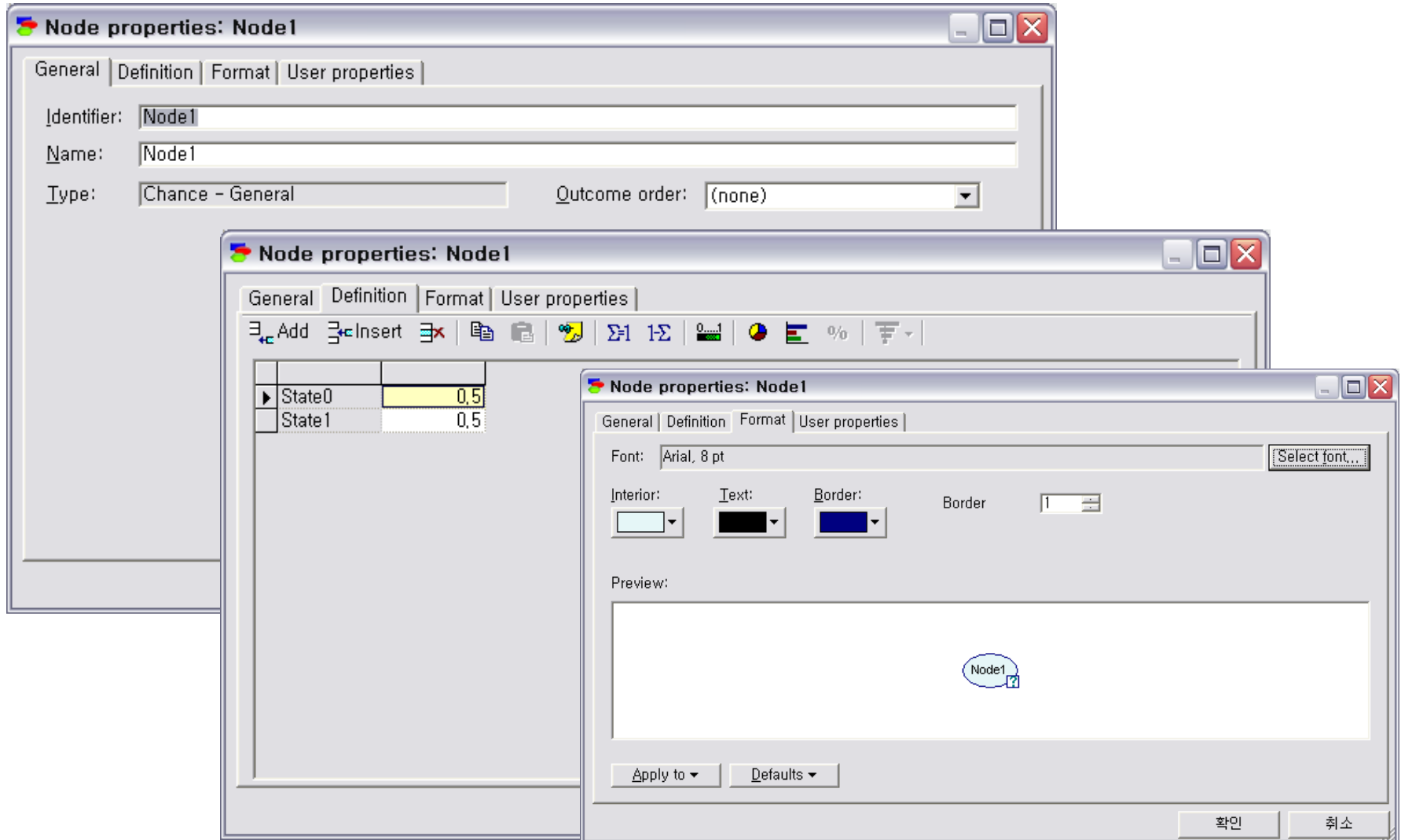
Practice 1: Building a Bayesian Network



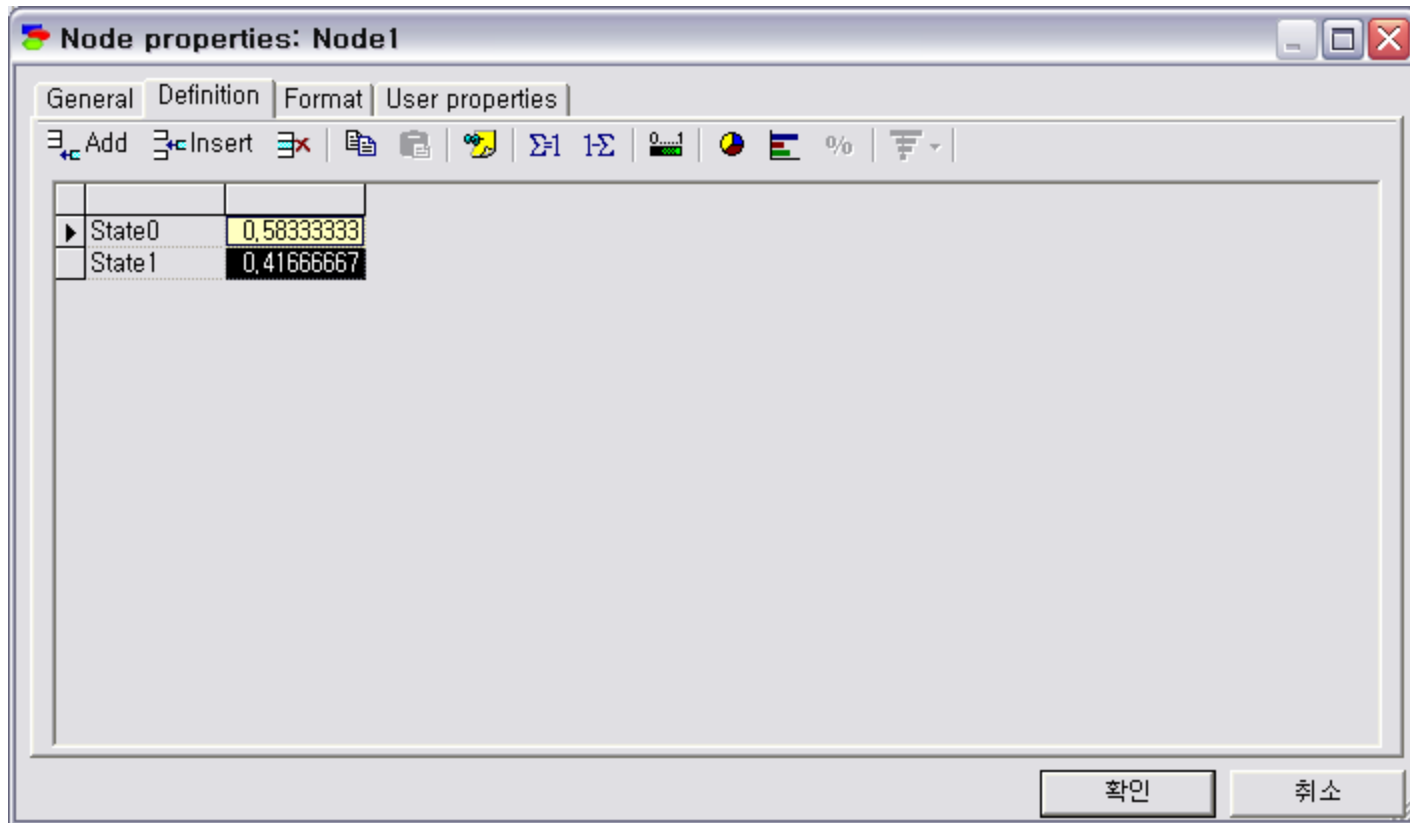
Practice 1: Building a Bayesian Network



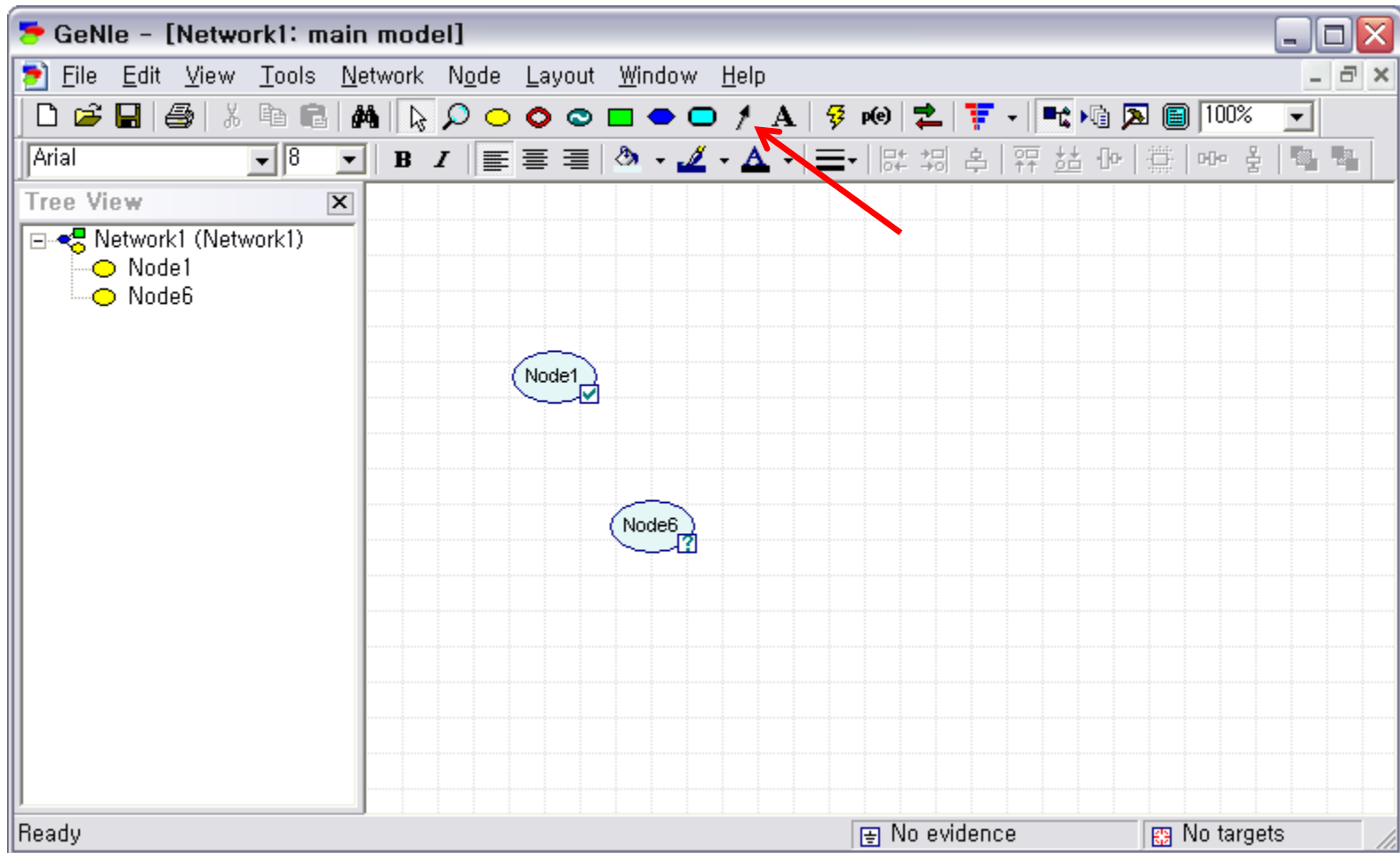
Practice 1: Building a Bayesian Network



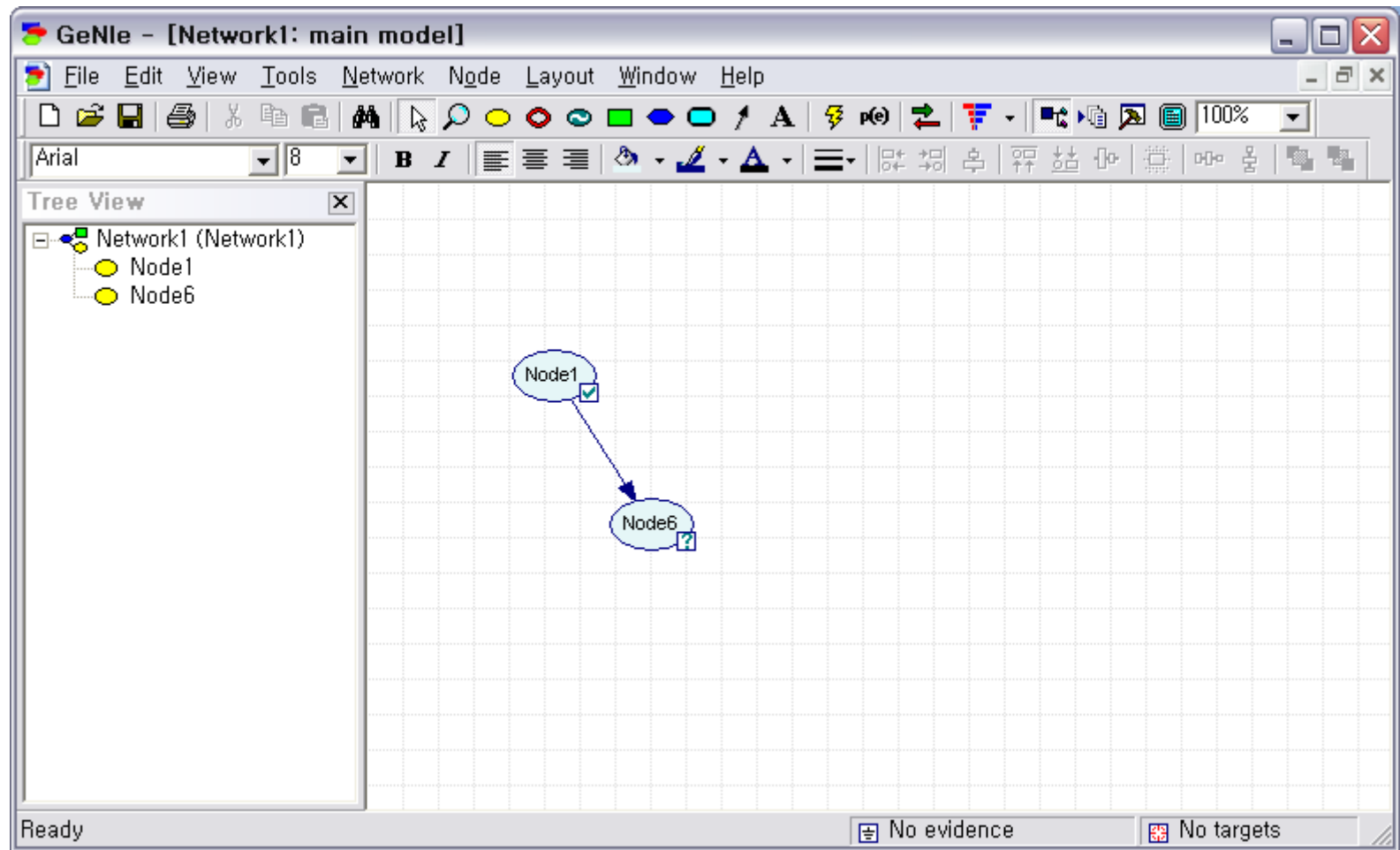
Practice 1: Building a Bayesian Network



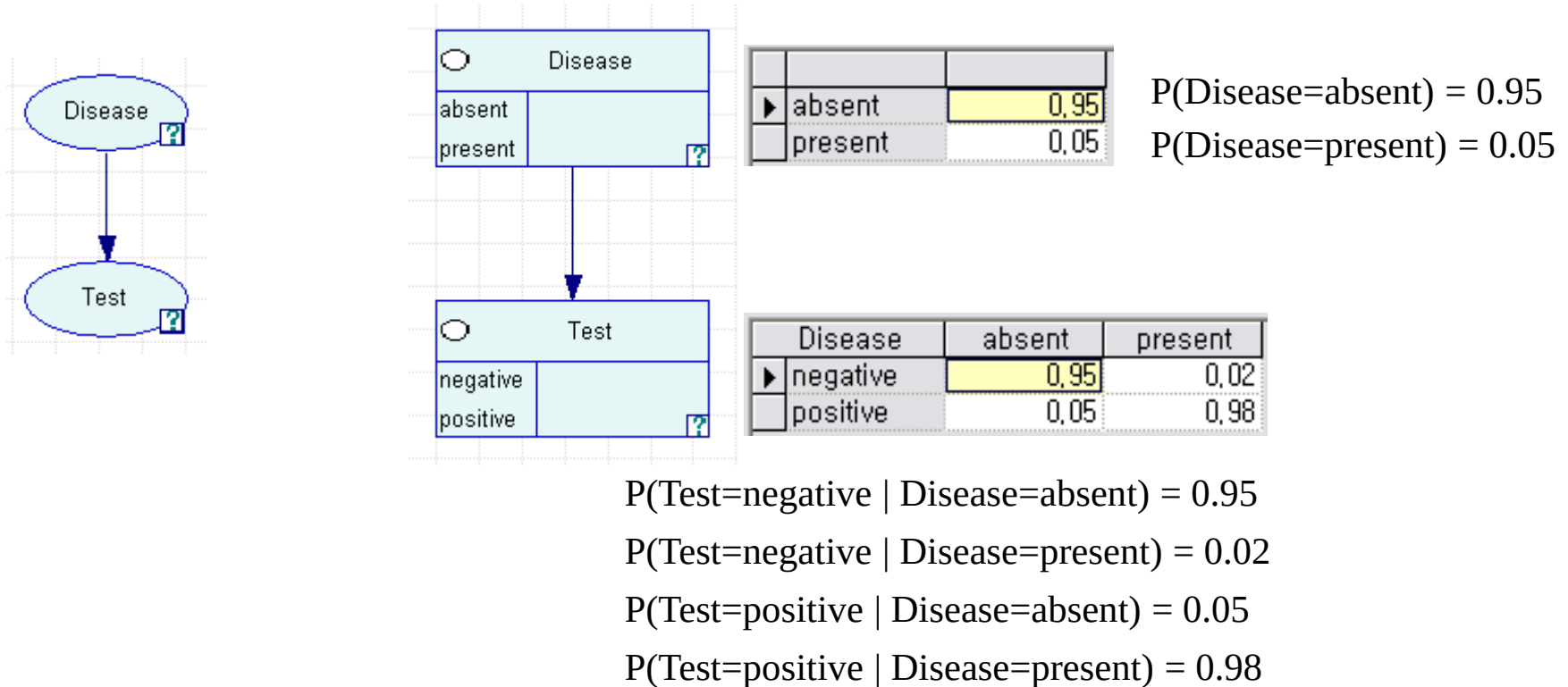
Practice 1: Building a Bayesian Network



Practice 1: Building a Bayesian Network

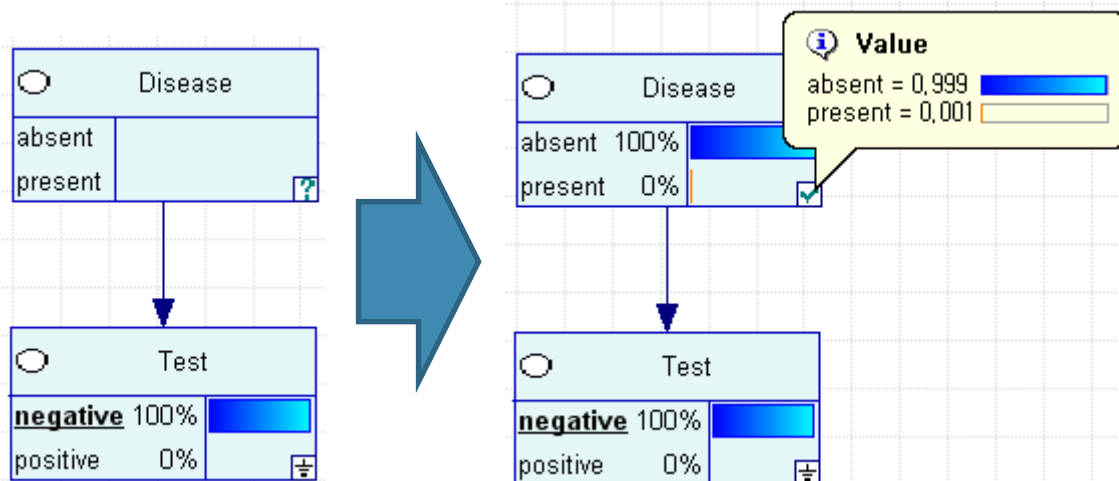
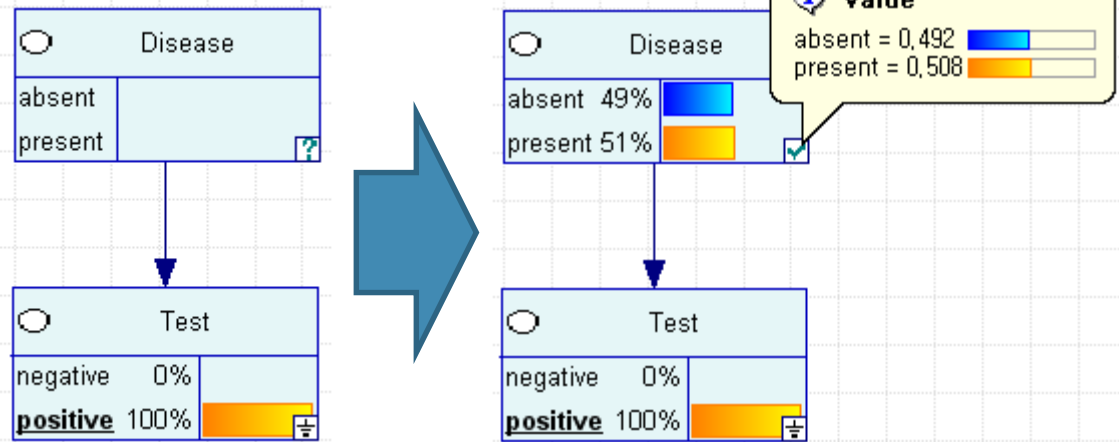


Practice 1: Disease-Test Model



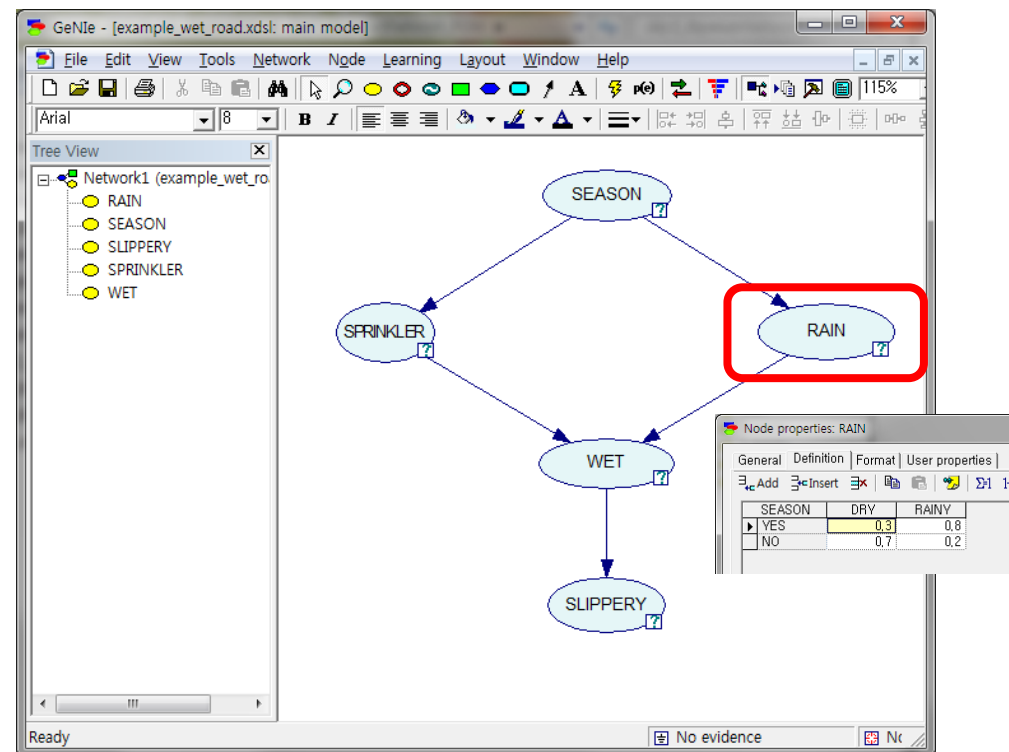
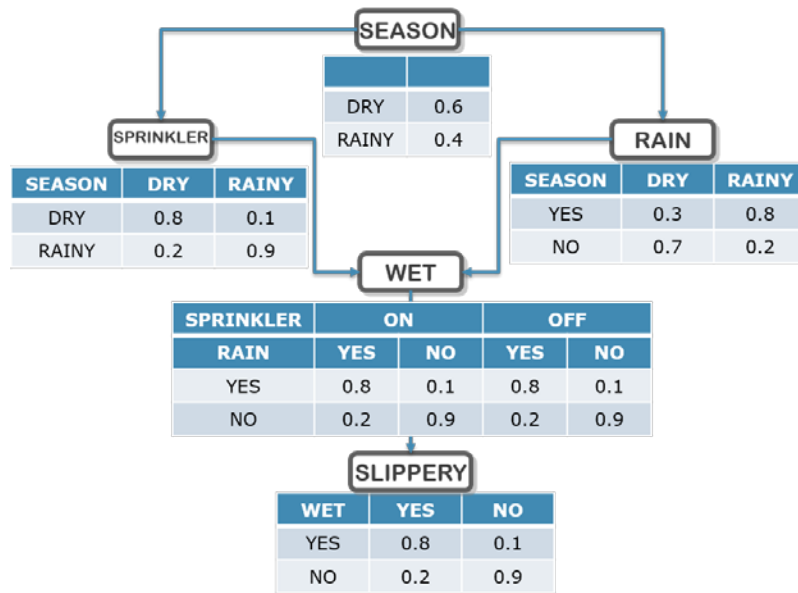
❖ Inference

- Set evidence
- Update belief
- Check updated probability



Practice 2: Inference with the Wet Road Model

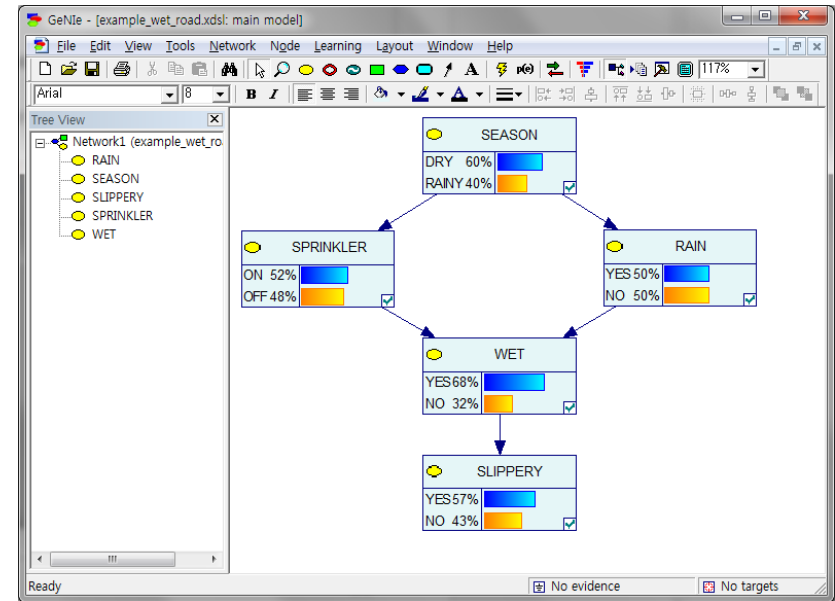
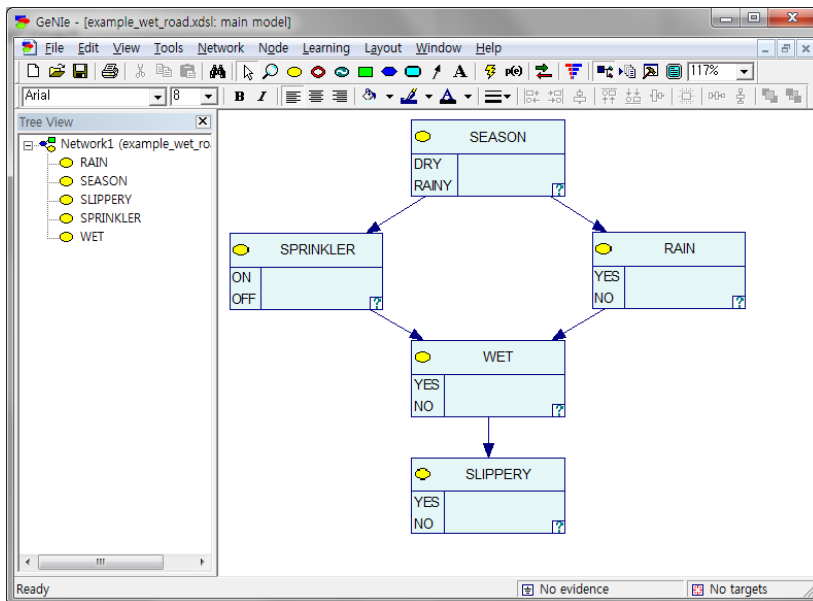
- ❖ Open '*example_wet_road.xdsl*' in GeNIe
- ❖ Check the structure
- ❖ Check the parameters
 - Double-click each node and choose the 'Definition' tab to see the 'conditional probability table (CPT)'



Practice 2: Inference with the Wet Road Model

❖ The first inference

- Represent each node as a bar chart
 - Check following menu: [Node – View as – Bar Chart]
 - Or just press 'F8'
- Inference: get marginalized probabilities for all the nodes
 - Check following menu: [Network – Update Beliefs]
 - Or just press 'F5'



Practice 2: Inference with the Wet Road Model

❖ More basic inferences for Q&A's

- Q1: If the season is dry and the pavement is slippery, did it rain?

❖ Analysis of the question

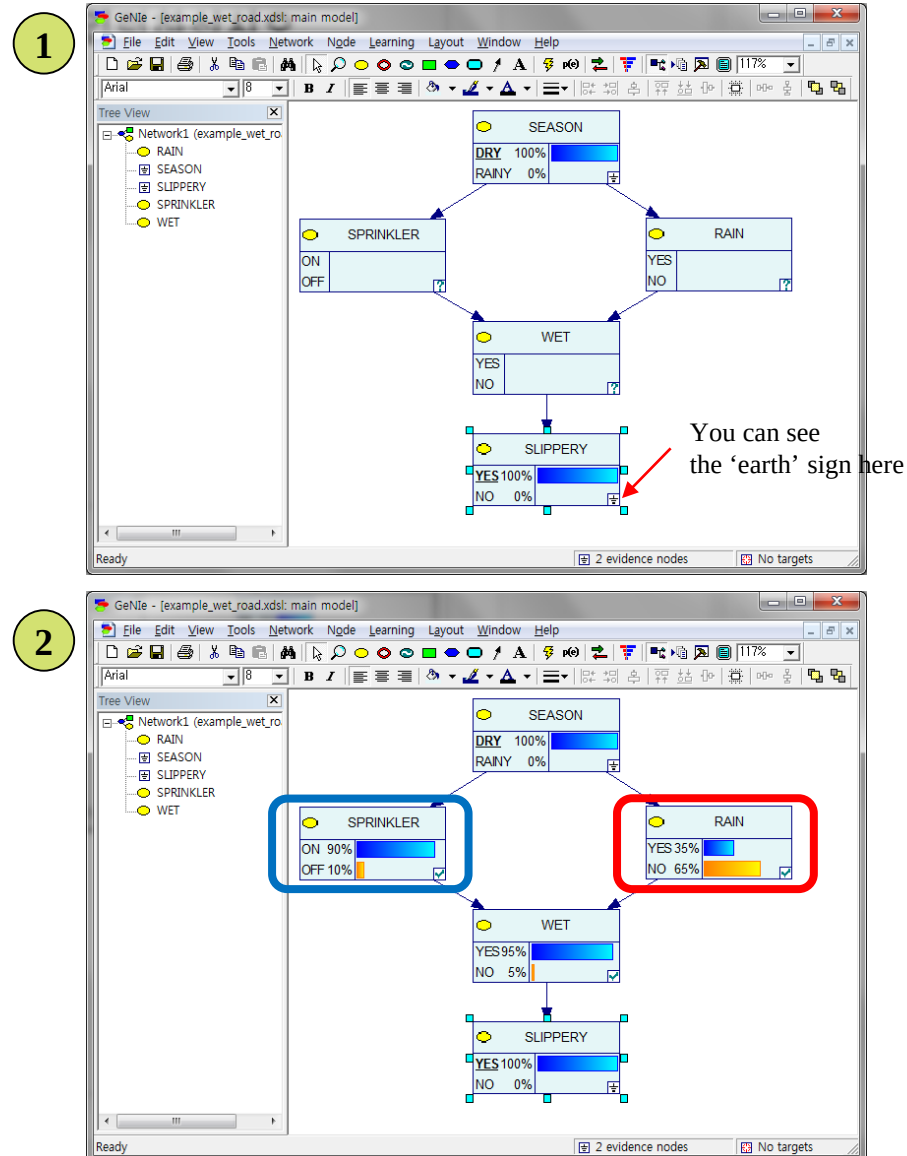
- Given evidences: SEASON = DRY, SLIPPERY = YES
- Target node to check: RAIN

❖ Inference

- 1 Set evidences by clicking the given values
- 2 Press 'F5' to update beliefs and check the updated bar chart of the target and related nodes

❖ Analysis of the inference result

- A1: Unlikely, it is more likely that the sprinkler was ON with a very slight possibility that it is not even wet.



Practice 2: Inference with the Wet Road Model

❖ More basic inferences for Q&A's

- Q2: But what if we see that the sprinkler is OFF?

❖ Analysis of the question

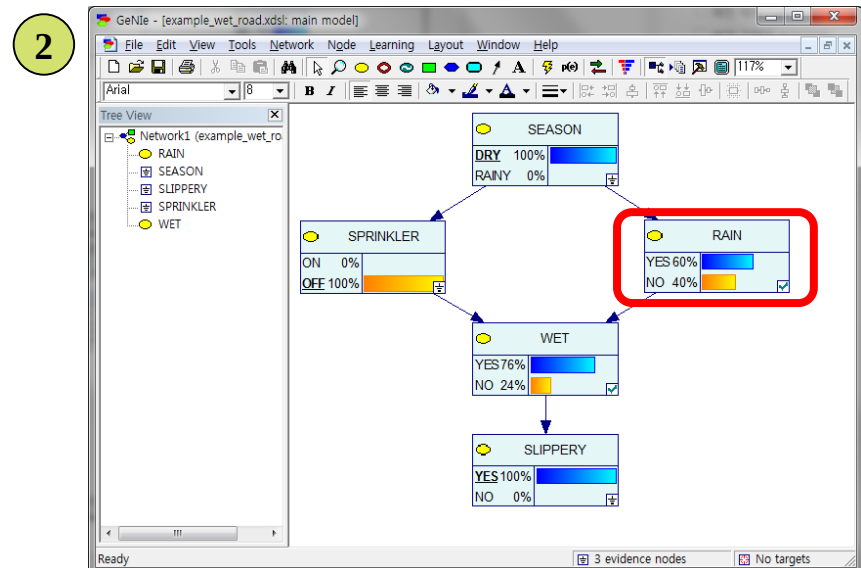
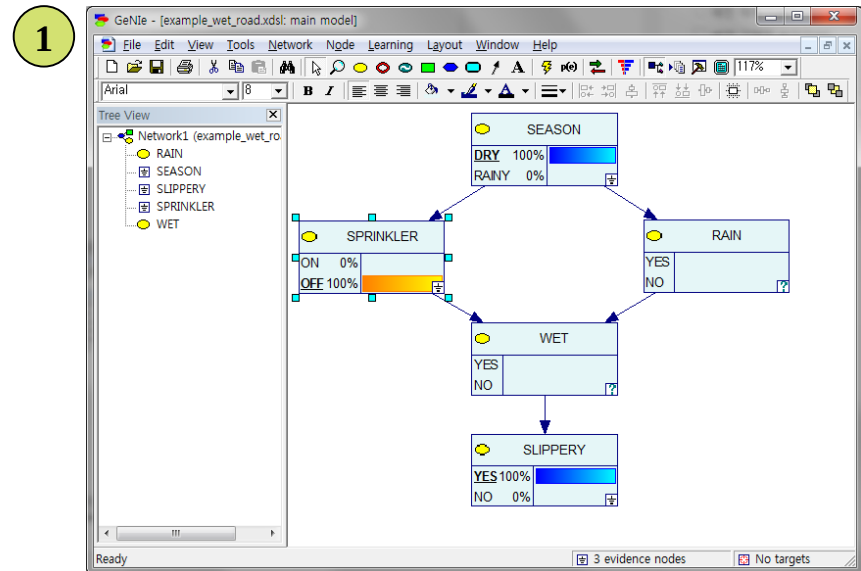
- More evidence: SPRINKLER = OFF
- Target node to check: RAIN

❖ Inference

- 1 Set evidences by clicking the given values
- 2 Press 'F5' to update beliefs and check the updated bar chart of the target node

❖ Analysis of the inference result

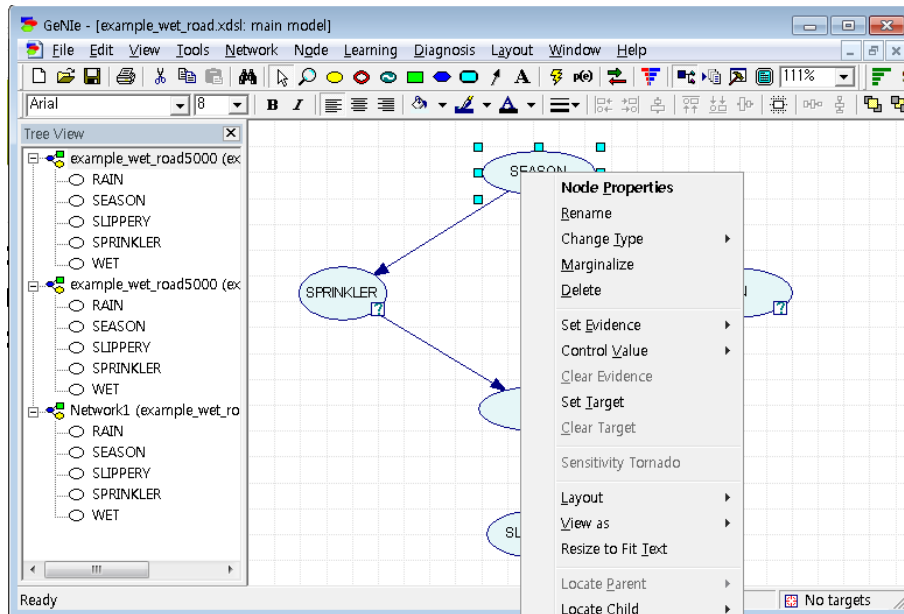
- A2: Then it is more likely that it rained



Practice 2: Inference with the Wet Road Model

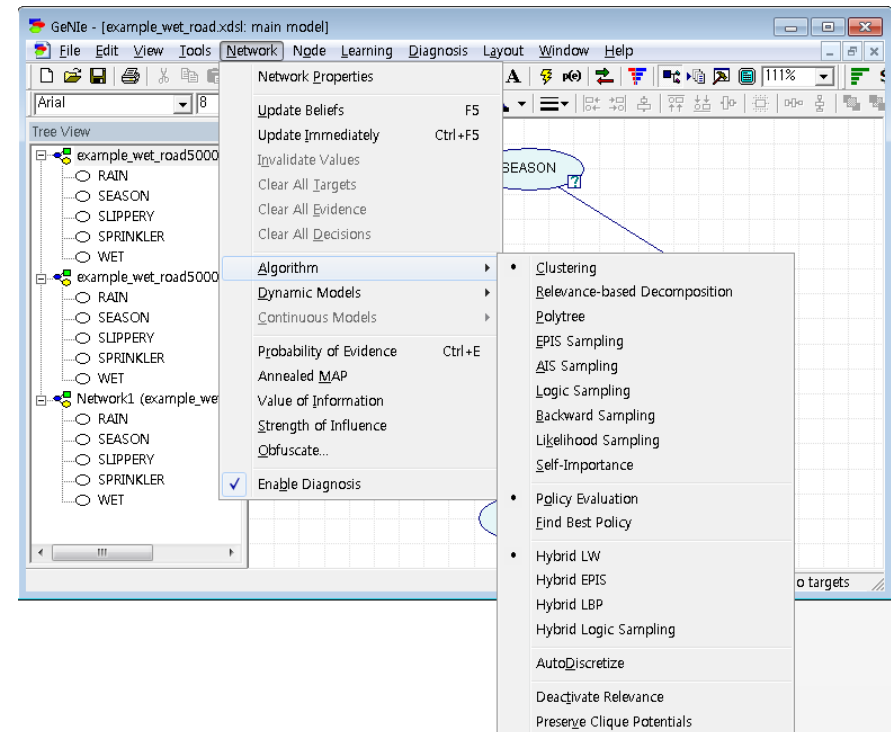
❖ For each node, we can set following options for various inference

- Set/clear target
- Set/clear evidence
- Control Value



❖ Algorithms for inference in Bayesian networks

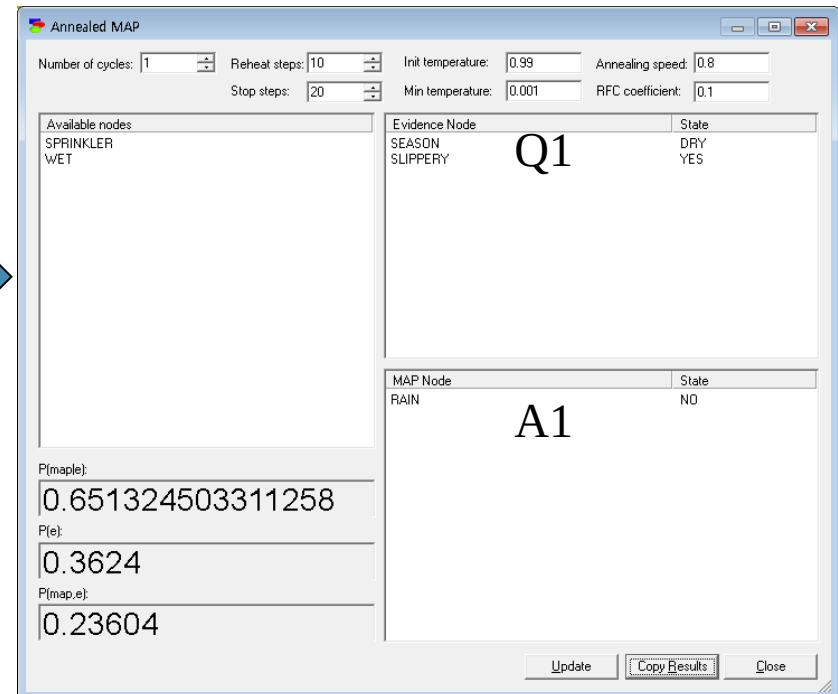
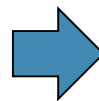
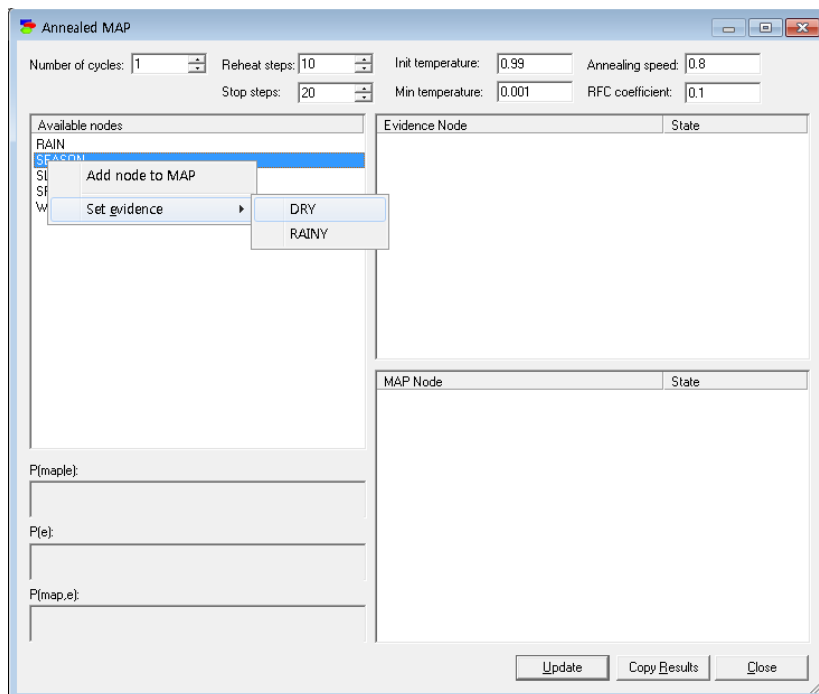
- Exact methods
- Approximate methods



Practice 2: Inference with the Wet Road Model

❖ Inference: Probability of Evidence, Annealed MAP

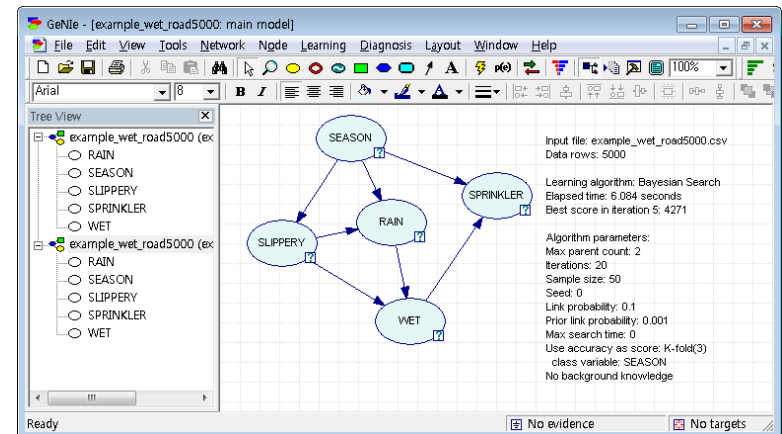
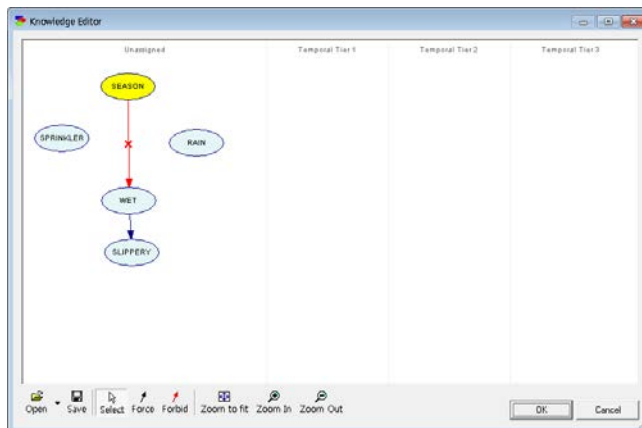
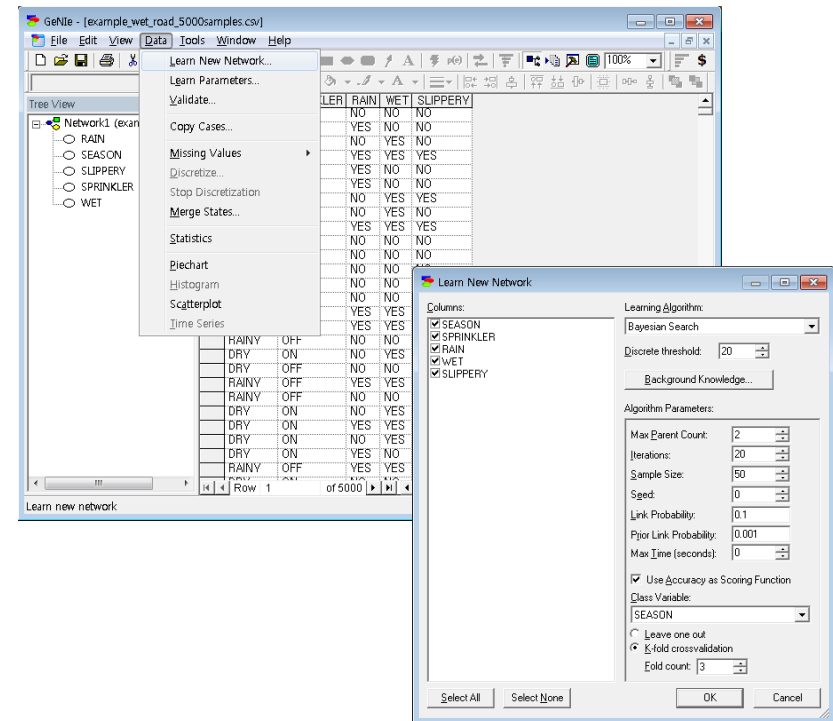
- Based on a set of observed nodes e
 - We can calculate the probability of this observed configuration
 - We can estimate the most probable states of target nodes X
- Menu: [Network – Probability of Evidence, Annealed MAP]
 - $P(e)$: probability of evidence
 - MAP(maximum a posteriori): maximizing the posterior $P(X | e)$



Practice 3: Learning from Data

❖ Learning Bayesian networks using GeNIe

1. In GeNIe, open 'example_wet_road_5000samples.csv'
2. [Data-Learn New Network]
3. Setting parameters for learning
 1. Learning Algorithm: Bayesian Search
 2. Max Parent Count: 2
 3. Use Accuracy as Score Function. Class Variable: SEASON. 3-fold cross validation
4. Prior, background knowledge can be applied by forcing or forbidding edges
5. To apply other learning algorithms, go to the data matrix by selecting corresponding window in [Window] menu



Practice 3: Learning from Data

❖ Learning Bayesian networks using Weka

1. In Weka Explorer, open 'example_wet_road_5000samples.csv'
2. In the Classify tab, choose [weka.classifiers.bayes.BayesNet]
3. Set following parameters for BayesNet
 1. SearchAlgorithm: K2
 2. initAsNaiveBayes: False
 3. maxNrOfParents: 2
4. Set 'Use training set' as the test option
5. Set 'SEASON' as the class label (usually located at the root node)
6. Click the 'Start' button
7. At the generated item in 'Result list', right-click and choose 'visualize graph'
8. Ignore the contents in 'Classifier output'
9. Click each node to check parameters

You can see that the structure and parameters of the original BN were restored by learning

The image displays three screenshots from the Weka software interface. The top-left screenshot shows the 'weka.gui.GenericObjectEditor' window for 'weka.classifiers.bayes.BayesNet', with parameters like 'searchAlgorithm' set to 'K2 -P 2 -N -S BAYES'. The top-right screenshot shows the 'weka.gui.GenericObjectEditor' window for 'weka.classifiers.bayes.net.search.local.K2', with parameters like 'initAsNaiveBayes' set to 'False' and 'maxNrOfParents' set to '2'. The bottom-left screenshot shows the 'Weka Explorer' window with the 'Classify' tab selected, displaying the 'BayesNet' classifier and its output, including a 'Confusion Matrix' and 'Detailed Accuracy By Class'. The bottom-right screenshot shows the 'Weka Classifier Graph' window, displaying a Bayesian network structure with nodes 'SEASON', 'SPRINKLER', 'RAIN', 'WET', and 'SLIPPERY'.

Weka Explorer Classifier output:

Classifier	Root relative squared error	Coverage of cases (0.95 level)	Mean rel. region size (0.95 level)	Total Number of Instances
BayesNet -D -Q weka.classifiers.bayes.net.search.local.K2 -- -P 2 -N -S BAYES -E weka.classifiers.bayes.net.estr	75.6191 %	99.42 %	96.64 %	5000

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall
0.688	0.057	0.901	0.688	
0.943	0.312	0.800	0.943	
Weighted Avg.	0.833	0.203	0.843	0.833

=== Confusion Matrix ===

a \ b	a = NO	a = YES
b = NO	1479	672
b = YES	163	2686

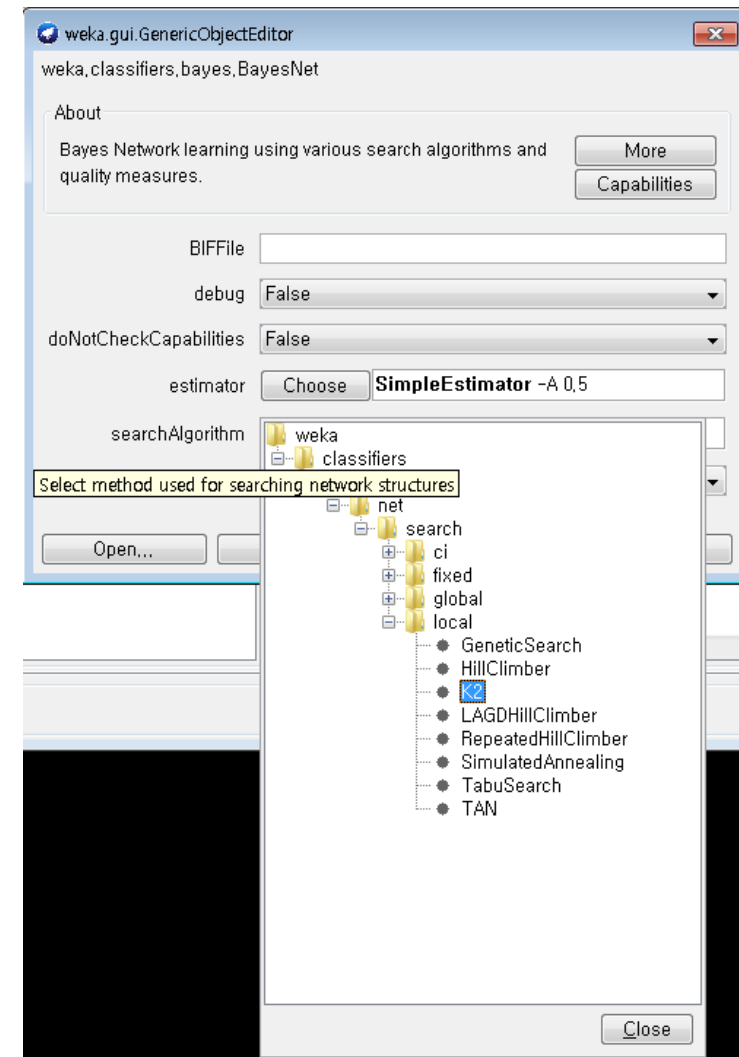
Weka Classifier Graph structure:

```
graph TD; SEASON --> SPRINKLER; SEASON --> RAIN; SPRINKLER --> WET; RAIN --> WET; WET --> SLIPPERY
```

Practice 3: Learning from Data

❖ Learning Bayesian networks using Weka

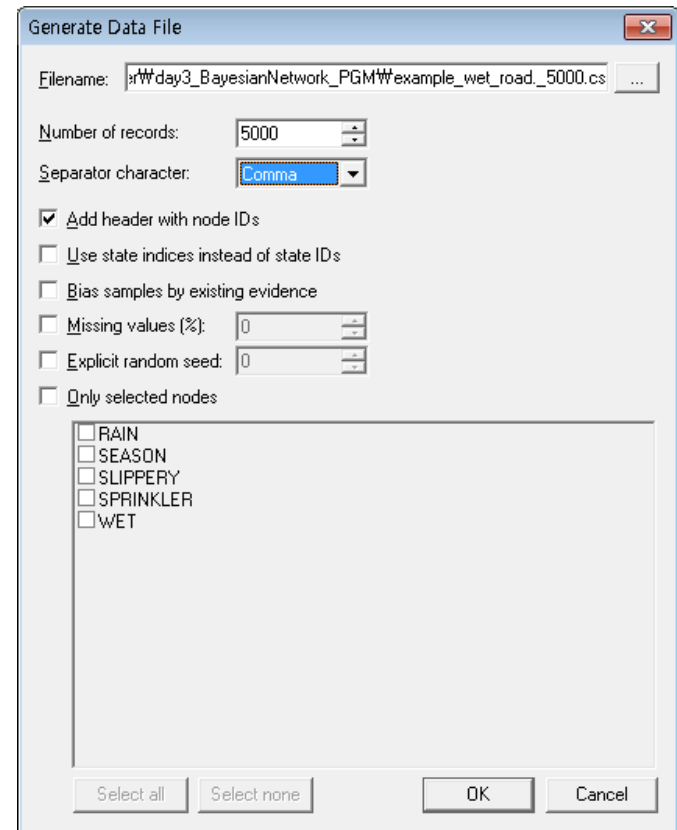
- You may try various search algorithms
 - GeneticSearch
 - HillClimber
 - K2: HillClimber with the fixed order of variables
 - ..
- You may try various score metrics
 - BAYES
 - BDeu
 - MDL
 - Entropy
 - AIC
- You may try various estimators and alpha values



Practice 3: Learning from Data

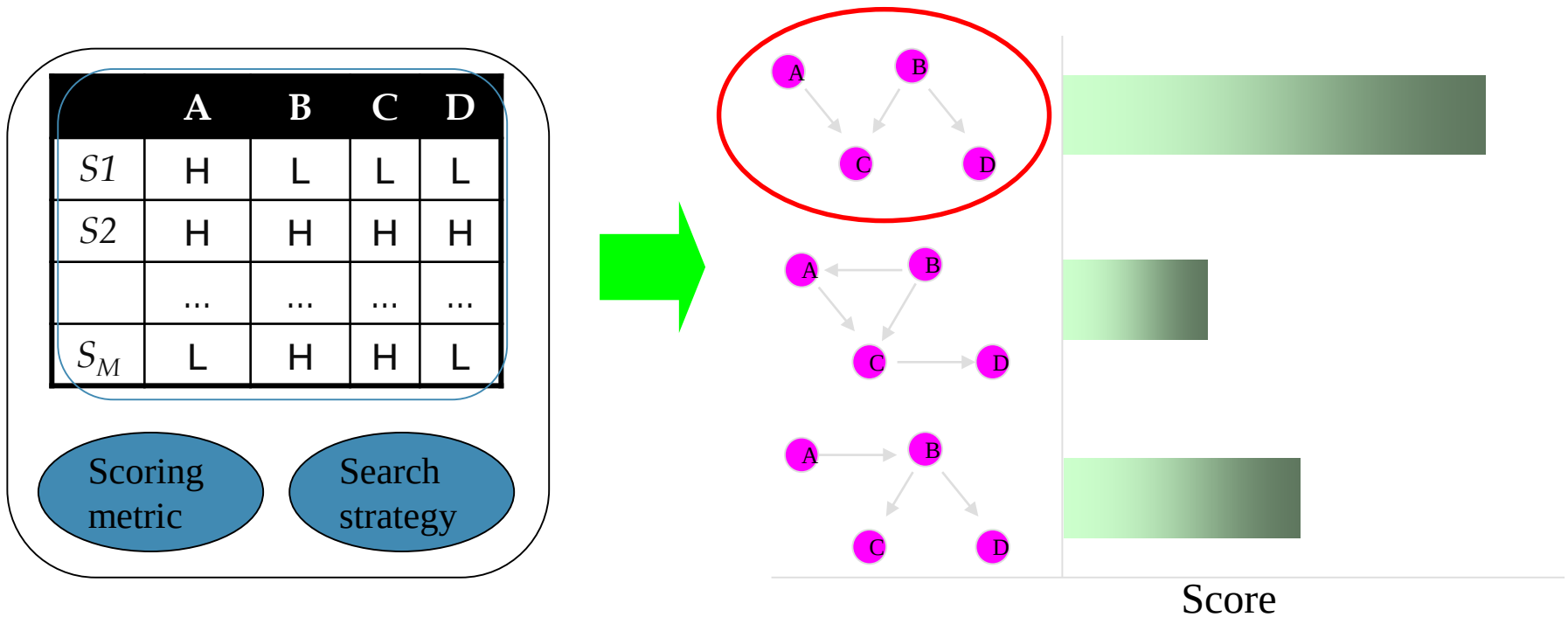
❖ We can generate samples that follow a Bayesian network model

1. Load a Bayesian network model
2. [Learning – Generate Data File]
 - example_wet_road_5000samples.csv is a generated file
 - simulation of the wet road model
 - 5,000 instances are sampled from the wet-road model



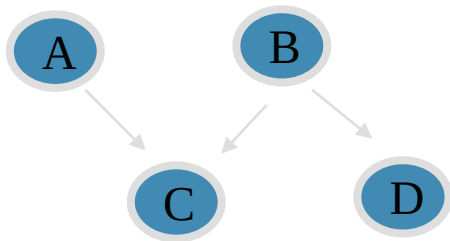
Understanding Structure Learning (Score-Based Approaches)

- ❖ Approach the problem as an optimization problem
- ❖ Use a scoring metric to measure how well a particular structure fits the observed set of cases



Understanding Parameter Learning

❖ **Task: Given a network structure, estimate the parameters of the model from data.**



	A	B	C	D
S_1	H	L	L	L
S_2	H	H	H	H
	...	..	...	...
S_M	L	H	H	L



P(A)	
H	L
0.99	0.01

P(B)	
H	L
0.93	0.07

P(C)		
(A, B)	H	L
(H, H)	0.4	0.6
(H, L)	0.2	0.8
(L, H)	0.3	0.7
(L, L)	0.8	0.2

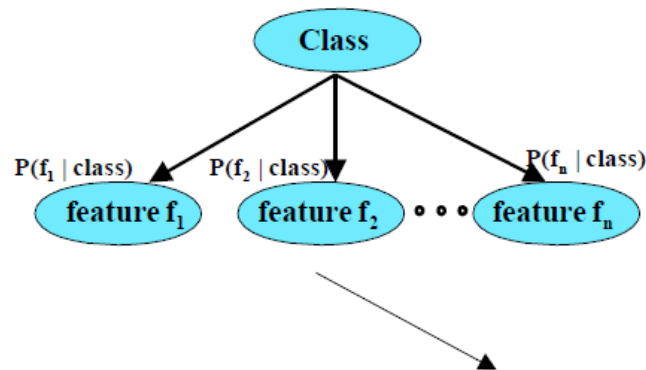
P(D)		
B	H	L
H	0.9	0.1
L	0.1	0.9

Structures of Bayesian Networks

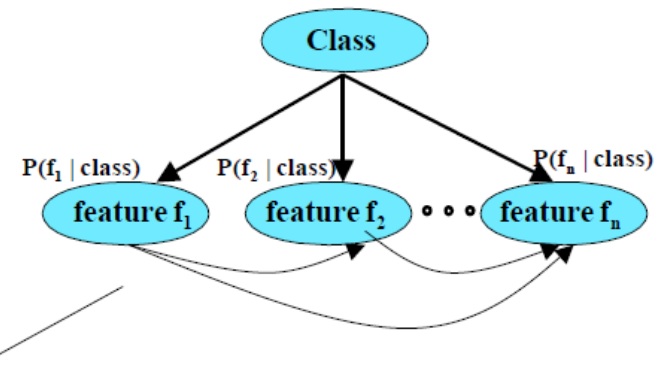
Slide from I.Rish, 'A Tutorial on Inference and Learning in Bayesian Networks'

Model selection trade-offs

Naïve Bayes – too simple
(less parameters, but bad model)



Unrestricted BN – too complex
(possible overfitting + complexity)

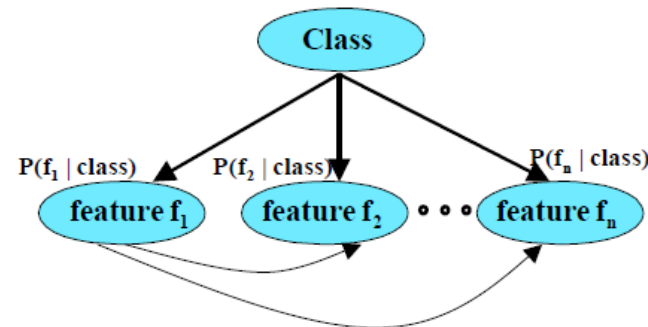


Various approximations between the two extremes

TAN:

tree-augmented Naïve Bayes
[Friedman et al. 1997]

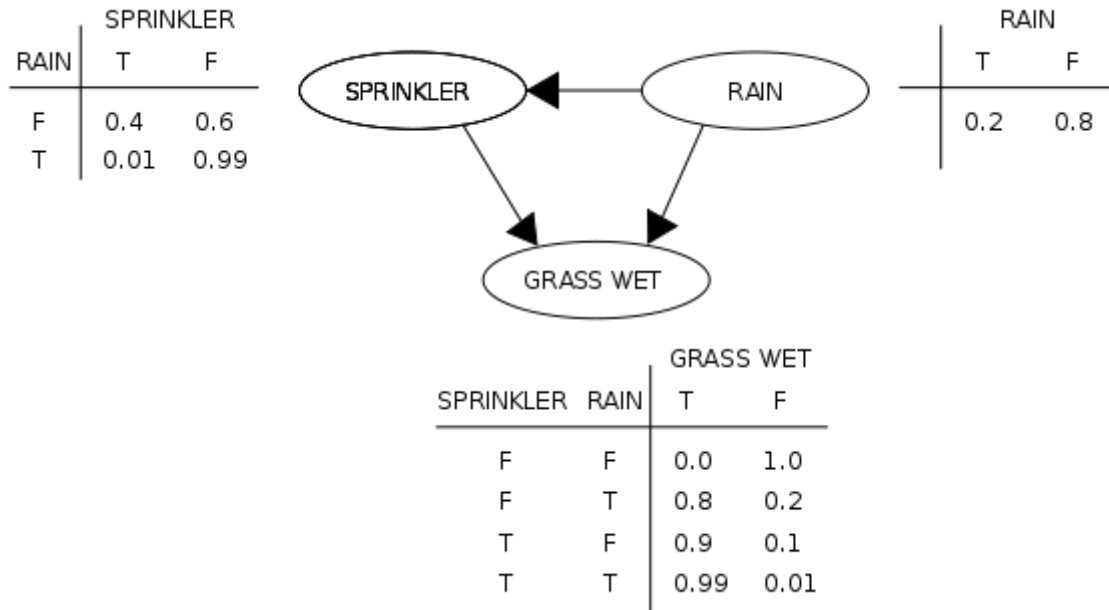
Based on Chow-Liu Tree Method
(CL) for learning trees
[Chow-Liu, 1968]





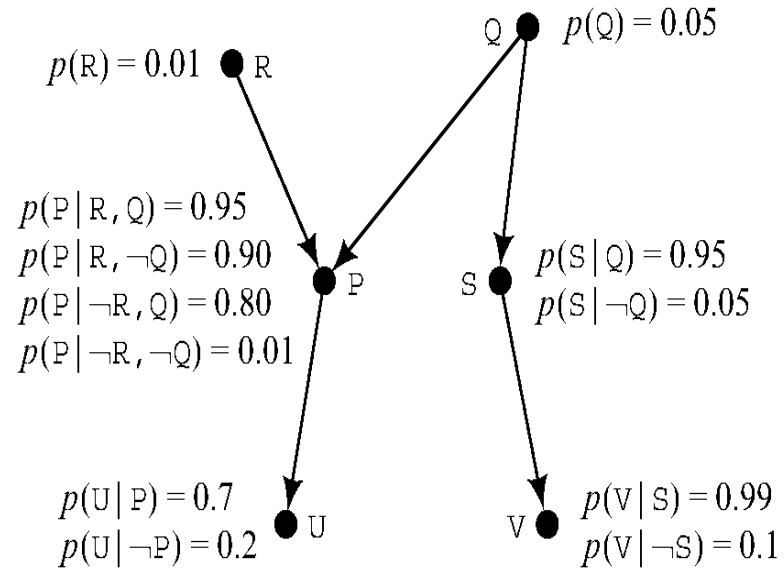
Homework and Exercises

Homework 1 (inference)



What is the probability that it is raining, given the grass is wet?

Homework 2 (inference)



❖ Q1) $p(U|R,Q,S) = ?$

❖ Q2) $p(P|Q) = ?$

❖ Q3) $p(Q|P) = ?$

❖ First, you may try to calculate by hand

❖ Next, you can check the answer with GeNIe

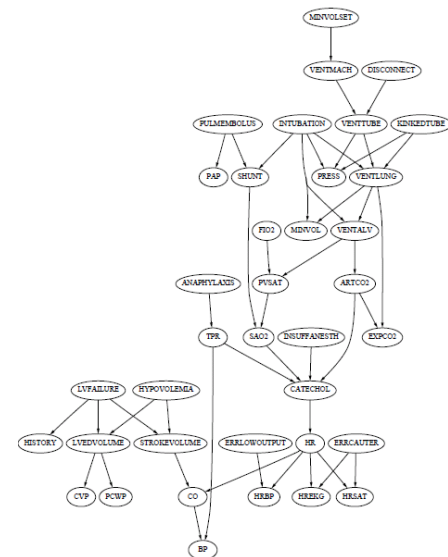
Dataset #1: Alarm Network

❖ Description

- The network for a medical diagnostic system developed for on-line monitoring of patients in intensive care units
- You will learn how to do **inference** with a given Bayesian network

❖ Configuration of the data set

- 37 variables, discrete (2~4 levels)
- Variables represent various states of heart, and lung
- Three kinds of variables
 - Diagnostic: basis of alarm
 - Measurement: observations
 - Intermediate: states of a patient



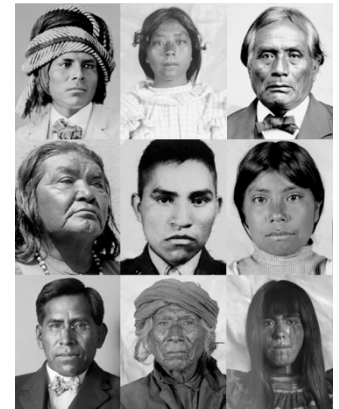
Dataset #2: Pima Indians Diabetes

❖ Description

- Pima Indians have the highest prevalence of **diabetes** in the world
- You will learn how to **learn structures and parameters** of Bayesian networks from data
- We may get **possible causal relationship** between features that affect diabetes in Pima tribe

❖ Configuration of the data set

- 768 instances
- 8 attributes
 - age, number of times pregnant, results of medical tests/analysis
 - **discretized set** will be used for BN
- Class value = 1 (Positive example)
 - Interpreted as "tested positive for diabetes"
 - 500 instances
- Class value = 0 (Negative example)
 - 268 instances



Dataset for Exercise with GeNIe

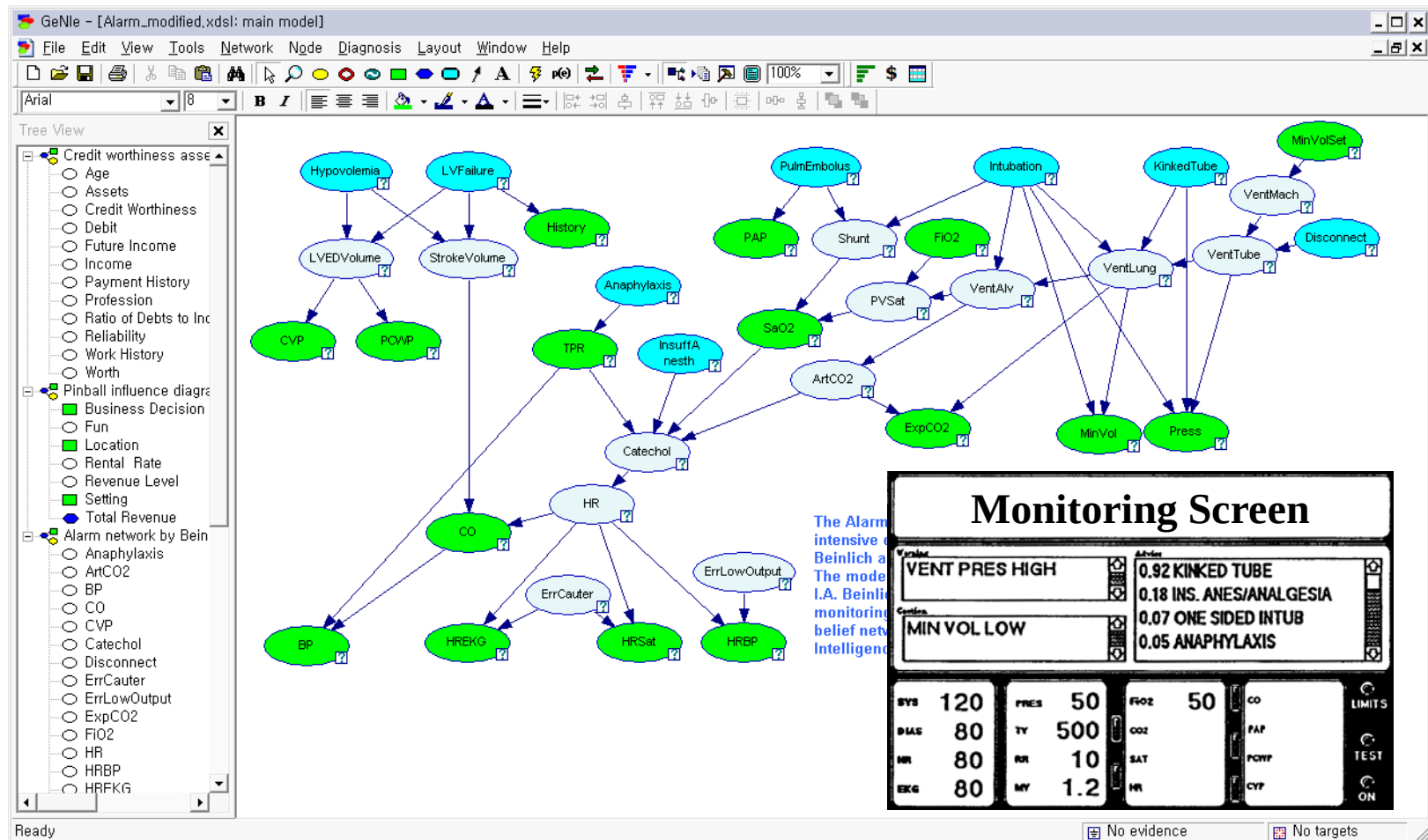
❖ Alarm Network

- data_Alarm_modified.xdsl

❖ Pima Indians Diabetes

- discretization with Weka: pima_diabetes.arff
(result: pima_diabetes_supervised_discretized.csv)
- Learning Bayesian network from data:
 - pima_diabetes_supervised_discretized.csv

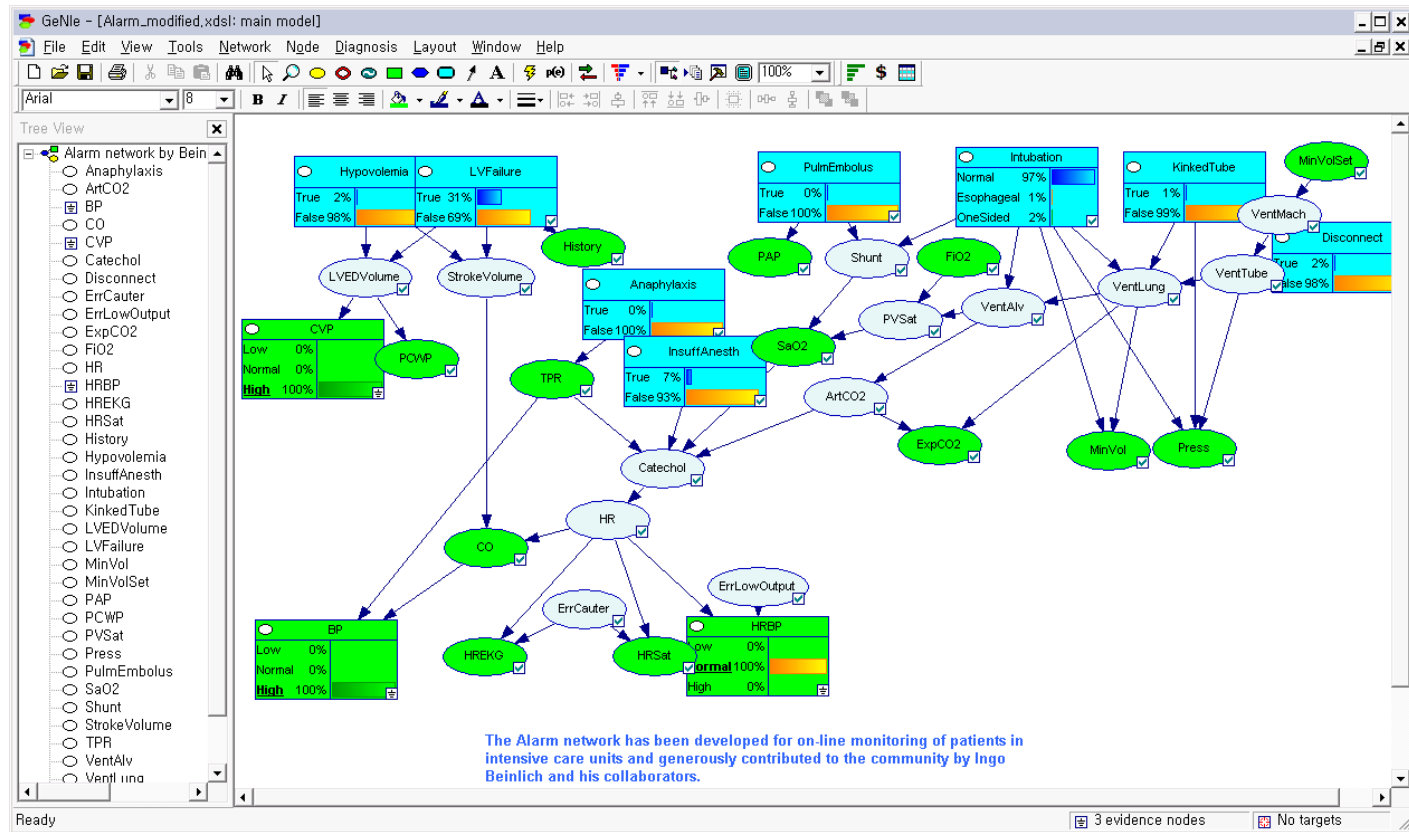
Exercise: Inference with the Alarm Network



○: diagnostic node ○: measurement node ○: intermediate node

❖ Inference tasks

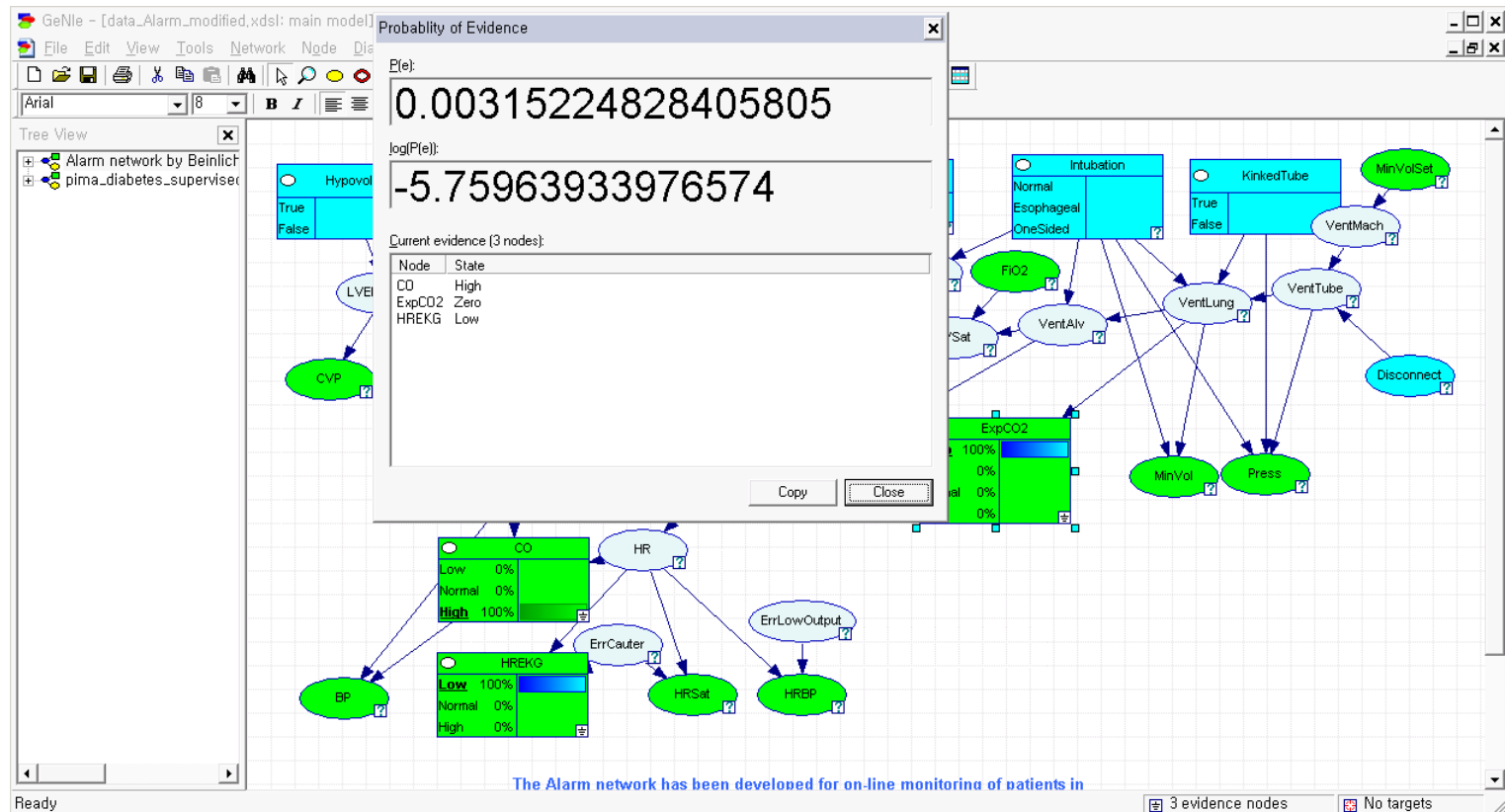
- Set evidences (according to observations or sensors)
- 'Network – Update Beliefs', or 'F5'



Exercise: Inference with the Alarm Network

❖ Inference tasks

- Network - Probability of Evidence



Exercise: Inference with the Alarm Network

❖ Inference tasks

- Based on a set of observed nodes
 - we can estimate the most probable states of target nodes
 - We can calculate the probability of this configuration
- Network - Annealed MAP

Annealed MAP

Number of cycles: 1 Reheat steps: 10 Init temperature: 0.99 Annealing speed: 0.8
Stop steps: 20 Min temperature: 0.001 RFC coefficient: 0.1

Available nodes	Evidence Node	State
Anaphylaxis	CO	High
ArtCO2	ExpCO2	Zero
BP	HREKG	Low
Catechol		
CVP		
Disconnect		
ErrCauter		
ErrLowOutput		
FIO2		
History		
HR		
HRBP		
HRSat		
Hypovolemia		
InsuffAnesth		
Intubation		
KinkedTube		
LVEDVolume		
LVFailure		
MirVol		

MAP Node

MAP Node	State
----------	-------

P(maple):
P(e):
P(map,e):

Update Copy Results Close



Annealed MAP

Number of cycles: 1 Reheat steps: 10 Init temperature: 0.99 Annealing speed: 0.8
Stop steps: 20 Min temperature: 0.001 RFC coefficient: 0.1

Available nodes	Evidence Node	State
Anaphylaxis	CO	High
ArtCO2	ExpCO2	Zero
BP	HREKG	Low
Catechol		
CVP		
Disconnect		
ErrCauter		
ErrLowOutput		
FIO2		
History		
HR		
HRBP		
Hypovolemia		
InsuffAnesth		
Intubation		
LVEDVolume		
LVFailure		
MirVol		
PAP		

MAP Node

MAP Node	State
HRSat	High
KinkedTube	False

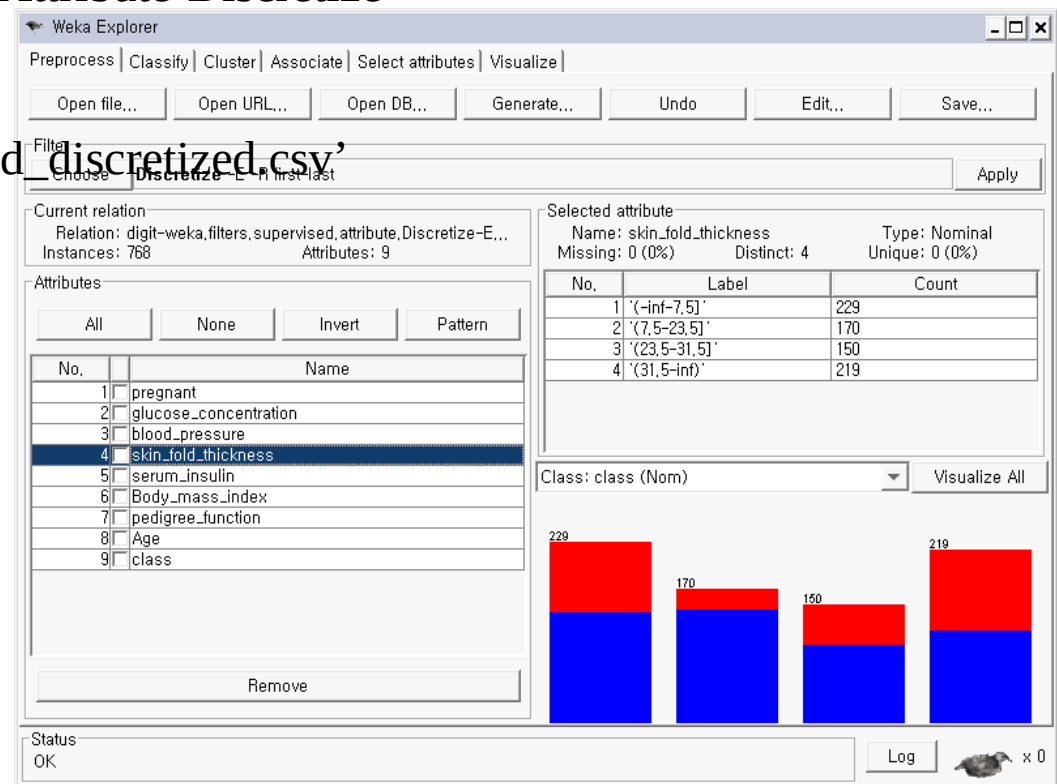
P(maple):
0.393291992418874
P(e):
0.00315224828405805
P(map,e):
0.00123975400823617

Update Copy Results Close

Exercise: Learning with Diabetes data

❖ Pima Indians Diabetes data

- Step 1: discretization of real-valued features with Weka
- 1. Open 'pima_diabetes.arff'
- 2. Apply 'Filter-Supervised-Attribute-Discretize' with default setting
- 3. Save into 'pima_diabetes_supervised_discretized.csv'



Exercise: Learning from Diabetes data

❖ Pima Indians Diabetes data

- Step 2: Learning structure of the Bayesian network
 1. File-Open Data File: pima_diabetes_supervised_discretized.csv
 2. Data-Learn New Network
 3. Set parameters as in Fig. 1
 4. Edit the resulting graph: changing position, color

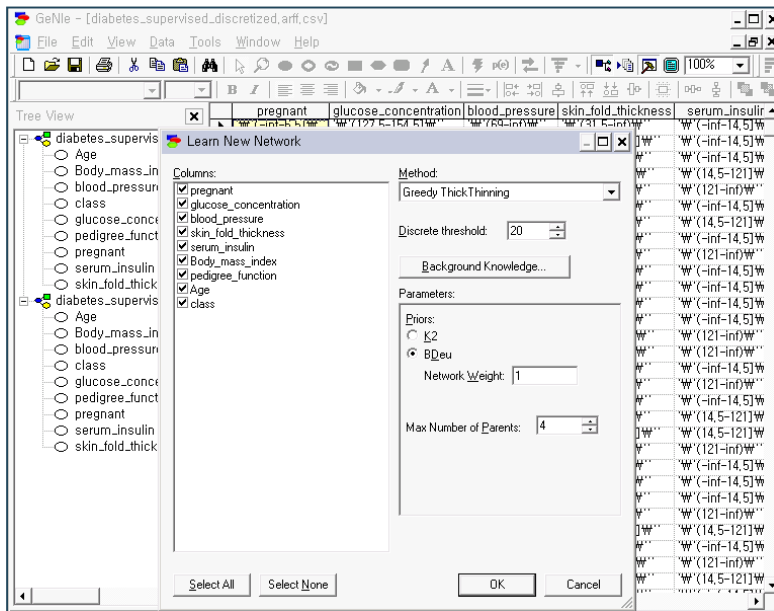


Fig. 1 Parameter setting

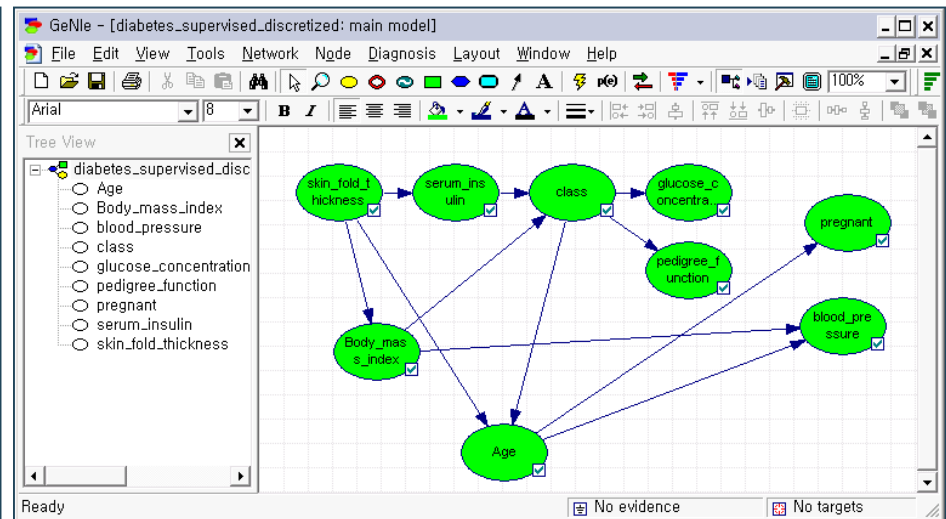


Fig. 2 Learned structure

Exercise: Learning from Diabetes data

❖ Pima Indians Diabetes data

■ Step 3: Learning parameters of the Bayesian network

1. Check the default parameters (based on counts in the data)
 1. 'F8' key will show distributions for all the nodes as bar chart
 2. 'F5' key will show you the probability
2. Network – Learn Parameters
3. Just click 'OK' button for each dialogue box
4. Check the change of the parameters with 'F5' key

